UWR Rainwater Offset Unit Standard (UWR RoU Standard)

Concept & Design: Universal Water Registry
www.uwaterregistry.io

Project Concept Note & Monitoring Report (PCNMR)

Project Name: Water Credit Project by NSL Tungabhadra Unit, Karnataka, India
UWR RoU Scope: 2 & 5
Monitoring Period: 01/01/2014 - 31/12/2025
Crediting Period[1]: 2014-2025
UNDP Human Development Indicator[2]: 0.644

[1] Not applicable. RoUs can be claimed for the "Lifetime of the Project Activity as per UWR RoU standard version 8.1
[2] Value is applicable for India according to latest HDI 2023-24 https://hdr.undp.org/sites/default/files/2023-24_HDR/HDR23-24_Statistical_Annex_HDI_Table.xlsx.
This is below 0.9 as per requirement mentioned in methodology rainwater offset methodology standard version 8.1.

A.1 Location of Project Activity

State	Karnataka
District	Bellary District
Block Basin/Sub Basin/Watershed	Please refer to http://cgwb.gov.in/watershed/basinsindia.html
Lat. & Longitude	15°39'37.94"N 76°52'15.96"E
Area Extent	141.97 acres (57.45 Ha)
No. of Villages/Towns	3; Desanur village

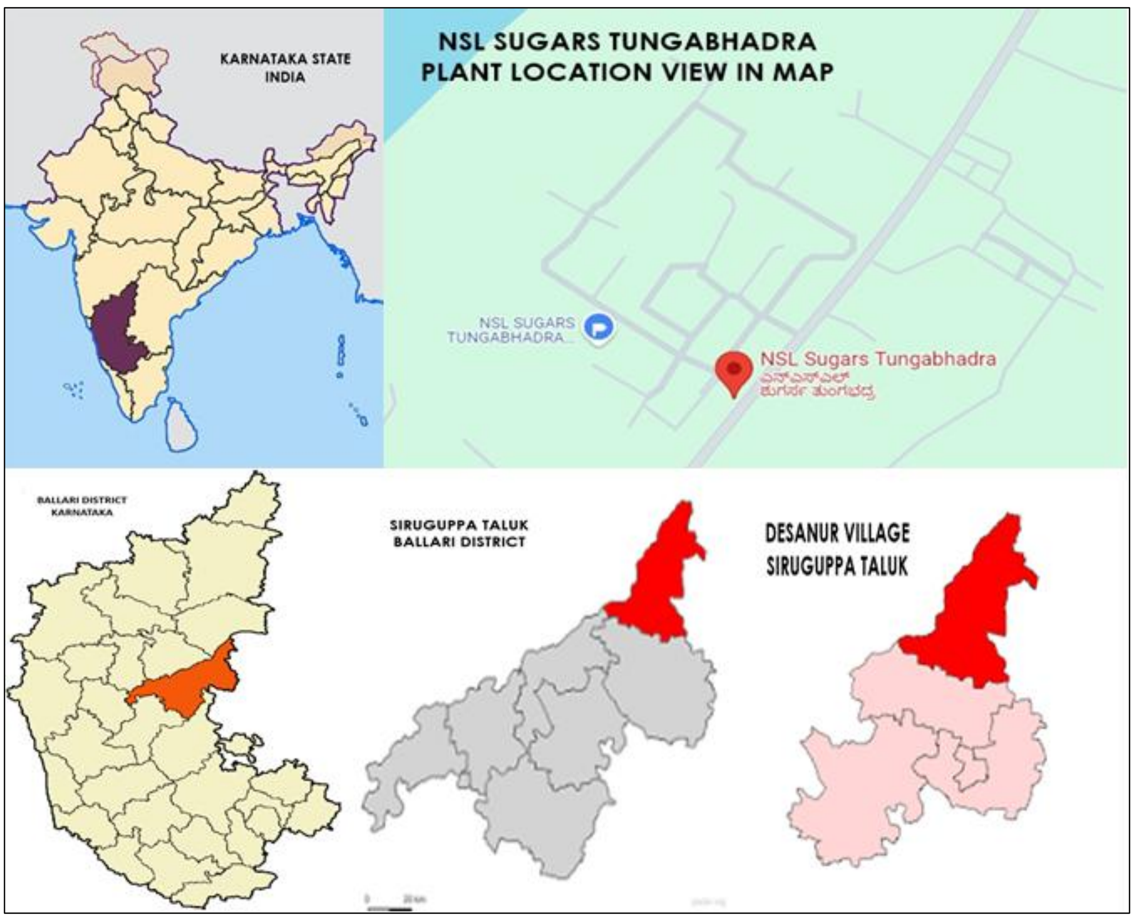


Figure 01: Location of NSL Sugars Tungabhadra ltd, Desanur Village, Siruguppa Tehsil, Ballari District, Karnataka State, India

Figure 02: Aerial view of NSL Sugars Tungabhadra ltd

A.2. Project owner information, key roles and responsibilities

NSL (Nuziveedu Seeds Limited) group foundation was led by Sri Venkata Ramaiah in 1973, and built upon by his son Sri M Prabhakar Rao, who has been leading the company since 1982. The company has evolved over the years with core expertise in research, marketing and supply chain to emerge as the largest seed company in India.

Name	Designation
Shri. M. Prabhakar Rao	Chairman of NSL Group
Shri Amarnath Reddy J	Sr. General Manager & Unit Head of NSL Sugars Limited, Tungabhadra

Roles and responsibilities are clearly documented here within the management plan including clear lines of accountability and reporting:-

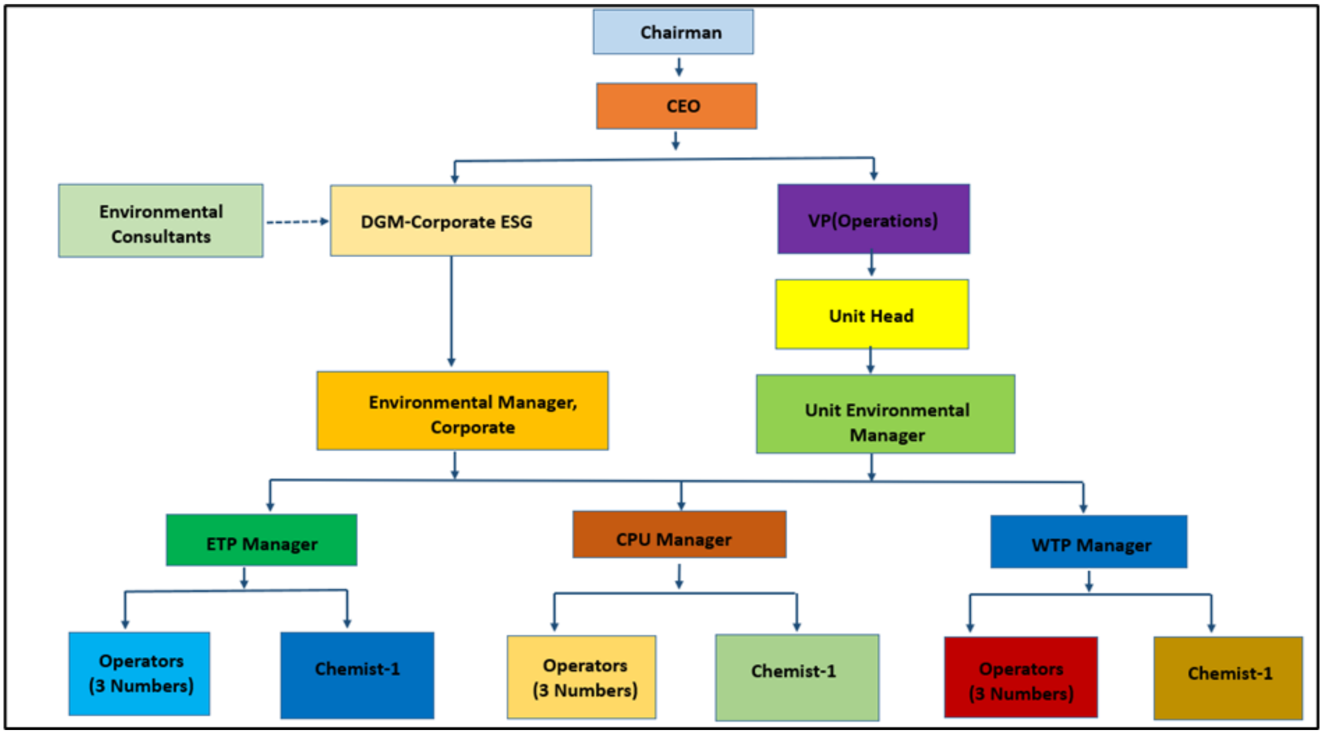


Figure 03: Environment Management Cell

S. No.	Designation	Responsibility
1	Chairman	• Strategic Oversight • Board Leadership • Compliance and risk management
2	CEO	• Allocate adequate resources, including budgets and personnel, to implement and maintain environmental initiatives. • Invest in technologies and infrastructure that support sustainable practices
3	DGM- Corporate ESG	Interact with regulatory authorities, local communities, and other stakeholders to address environmental concerns and promote sustainability initiatives.
4	VP (Operation)	Ensuring the alignment of environmental goals with operational objectives
5	Environmental Manager, Corporate	• Compliance Management • Environmental Monitoring and Reporting • Sustainability Initiatives • Training and Awareness • Incident management • Documentation and record keeping

6	Unit head	• Strategic Leadership • Compliance Oversight • Team management • Resource allocation • Policy Advocacy and Representation
7	Unit Environmental Manager	• Environmental Compliance • Develop strategies to mitigate adverse environmental impacts • Pollution Control • Resource management • Environmental Monitoring • Training and education
8	ETP Manager	• Resource management • Pollution Control • Effluent quality Monitoring • Training and education • To verify the data entered in log book regarding treated and untreated effluent quality
9	ETP chemist	• Effluent Analysis and Monitoring • Treatment Process Optimization
10	ETP Operators	Operation and Maintenance of ETP
11	CPU manager	• Pollution Control • To verify the data entered in log book regarding treated and untreated effluent quality • To verify the working efficiency of CPU
12	CPU Chemist	• Treated Effluent Analysis and Monitoring • Treatment Process Optimization
13	CPU operator	Operation and Maintenance of CPU
14	WTP Manager	Water quality analysis
15	WTP operator	Operation and Maintenance of WTP
16	WTP Chemist	• Water quality Analysis and Monitoring • Treatment Process Optimization
17	APC operator	Operation and Maintenance ESP
18	Supervisors	Development and Maintenance of Greenbelt and other general EMP of the industry

Table 01: Roles & responsibilities of Environment Management Cell at NSL Plant

The Project owner of NSL-TSL attest to the following:

(a) NSL Tungabhadra Unit owns the water user rights for the area within the project's boundary,

Consent For Operation
(CFO-Air,Water)

Consent No. AW-329847
Valid upto: 30/06/2026

Karnataka State Pollution Control Board
Parisara Bhavana,No.49, Church Street,Bengaluru-560001
Tele : 080-25589112/3, 25581383
Fax:080-25586321
email id: ho@kspcb.gov.in

Industry Colour: RED   Industry Scale: LARGE

(This document contains 5 pages including annexure & excluding additional conditions)

Combined Consent Order No. AW-329847   **PCB ID:** 11381   **Date:** 28/02/2022

Combined consent for discharge of effluents under the Water (Prevention and Control of Pollution) Act , 1974 and emission under the Air (Prevention and Control of Pollution)Act , 1981

Ref: 1. Application filed by the applicant/organization on 30/09/2021

2.Inspection of the Industry/organization/by RO, on 11/11/2021

3.Proceedings of the ECM dated 01/01/2022 ,held on 21/12/2021

Consent is hereby granted to the Occupier under Section 25(4) of the Water (Prevention & Control of Pollution) Act, 1974 (herein referred to as the Water Act) & Section 21 of Air (Prevention & Control of Pollution) Act, 1981, (herein referred to as the Air Act) and the Rules and Orders made there under and authorized the Occupier to operate /carryout industry/activity & to make discharge of the effluents & emissions confirming to the stipulated standards from the premises mentioned below and subject to the terms and conditions as detailed in the Schedule Annexed to this order.

Location:

Name of the Industry: Nsl Sugars (Tungabhadra) Ltd

Address: -, Desanur, Siruguppa.

Industrial Area: Not In I.A, Desanur,

Taluk: Siruguppa, District: Bellary

CONDITIONS:

a) Discharge of effluents under the Water Act:

Sr	Water Code	WC(KLD)	WWG(KLD)	Remark
1	Boiler Feed	400.000	135.000	Reused in Sugar unit and for cane farming after neutralization
2	Cooling Water	2338.000	400.000	Reused in Sugar unit and for cane farming instead of fresh water
3	Domestic Purpose	75.000	60.000	After treatement of water used for Gardening
4	Processing whereby water gets polluted and the pollutants are easily bio-degradable	0.000	350.000	After treatement of water used for Gardening & cane farming

b) Discharge of Air emissions under the Air Act from the following stacks etc.

Sl. No.	Description of chimney/outlet	Limits specified refer schedule
The details of Sources, control equipments and its specification, type of fuel,constituents to be controlled in emissions etc. are detailed in Annexure-II.		

The consent for operation is granted considering the following activities/Products;

Sr	Product Name	Applied Qty/Month	Unit
1	co-gen plant-28MW	28.0000	MWH
2	white crystel sugar with sugar cane crushing-3500TCD	105000.0000	TON

This consent is valid for the period from 01/07/2021 **to** 30/06/2026

To,

Nsl Sugars (Tungabhadra) Ltd

Desanur, Siruguppa.

CFO dated 28/02/2022 shows the amount of water usage permission from Government by NSL Tungabhadra unit

(b) NSL Tungabhadra Unit holds an uncontested legal land title for the project area within the project's boundary

ENVIRONMENTAL CLEARANCE

PARIVESH

(Pro-Active and Responsive Facilitation by Interactive, and Virtuous Environment Single-Window Hub)

Government of India
Ministry of Environment, Forest and Climate Change
(Impact Assessment Division)

To,

The General Manager
NSL SUGARS (TUNGABHADRA) LIMITED
NSL SUGARS (TUNGABHADRA) LIMITED, DESANUR, SIRUGUPPA TALUK, BELLARY KARNATAKA
583140,,Bellary,Karnataka-583140

Subject: Grant of Environmental Clearance (EC) to the proposed Project Activity under the provision of EIA Notification 2006-regarding

Sir/Madam,

This is in reference to your application for Environmental Clearance (EC) in respect of project submitted to the Ministry vide proposal number IA/KA/IND2/400621/2022 dated 04 Nov 2022. The particulars of the environmental clearance granted to the project are as below.

1.	**EC Identification No.**	**EC22A022KA110157**
2.	**File No.**	IA-J-11011/263/2022-IA-II(I)
3.	**Project Type**	New
4.	**Category**	A
5.	**Project/Activity including Schedule No.**	
6.	**Name of Project**	Establishment of 300 KLPD grain-based distillery plant and 15 MW captive power plant under EBP scheme by M/s NSL (Tungabhadra) Limited.
7.	**Name of Company/Organization**	NSL SUGARS (TUNGABHADRA) LIMITED
8.	**Location of Project**	Karnataka
9.	**TOR Date**	N/A

The project details along with terms and conditions are appended herewith from page no 2 onwards.

Date: 06/12/2022

(e-signed)
A N Singh
Scientist E
IA - (Industrial Projects - 2 sector)

Note: A valid environmental clearance shall be one that has EC identification number & E-Sign generated from PARIVESH.Please quote identification number in all future correspondence.

This is a computer generated cover page.

This has reference to your online proposal no. IA/KA/IND2/400621/2022 dated 04th November, 2022 for environmental clearance to the above-mentioned project.

2. The Ministry of Environment, Forest and Climate Change has examined the proposal seeking environmental clearance for establishment of 300 KLPD Grain based Ethanol Plant & 15 MW captive power plant (Bagasse/rice husk based) located at Village Desanur, Tehsil Siruguppa, District Bellary, State Karnataka by M/s. NSL Sugars (Tungabhadra) Limited. The grain-based distillery is proposed in the premises of existing sugar plant of 3500 TCD and 28 MW Cogeneration complex.

3. As per the MoEF&CC Notification S.O. 2339(E), dated 16th June, 2021, a special provision in the EIA Notification, 2006-(Schedule 5(ga), Category B2) is made, wherein for all applications made for Grain based distilleries with Zero Liquid Discharge producing Ethanol; solely to be used for Ethanol Blended Petrol Programme of the Government of India shall be considered under B2 Category and appraised at Central Level by Expert Appraisal Committee (EAC) with condition that the project proponent shall file a Notarized Affidavit that ethanol produced from proposed project shall be used completely for EBP Programme.

4. The details of products and capacity as under:

S. No.	Name of unit	Name of the product/ by-product	Total production capacity
1	Distillery (grain based)	Ethanol	300 KLPD
2	Co-generation power plant of distillery	Power	15 MW
3	DWGS	DDGS	219 TPD
4	Fermentation unit	Carbon di-oxide	150 TPD

5. SEIAA has issued Environmental Clearance to the existing Industry for a capacity of 3500 TCD Sugar plant and 28 MW Cogeneration unit vide File No. SEIAA: 56 IND: 2008 dated 14.08.2009. Certified Compliance report of existing EC has been obtained from Integrated Regional Office, MoEFCC, Bangalore vide File no-EP/12.1/SEIAA/221/KAR/1560 dated 7.03.2022. The IRO has reported the status of compliance of the project is rated as satisfactory. EAC was satisfied with the response of PP.

Above Photographs shows the Environmental clearance which clearly declares the land occupied and details of project implementation and capacity by the Koppa unit

(c) NSL Tungabhadra Unit holds all necessary permits to implement the project or has applied for the same.- Yes, Unit possess all necessary documents like Environmental Clearance from MoEF&CC and **CFO** from State Pollution Control Board as attached above.

(d) Project implementation Cost- The estimated project cost was referred as INR 17,58,69,743 /-

A.2.1 Project RoU Scope

PROJECT NAME	Water Credit Project by NSL Tungabhadra Unit, Karnataka, India
UWR Scope:	Scope-2 & 5
Date PCNMR Prepared	Current Version: 12/06/2026 Final Version: NA (will be considering during Verification)

A.3. Land use and Drainage Pattern

For Scope-5

Not Applicable, since this scope of project activity involves treating and reusing wastewater from distillery and sugar plant.

It doesn't include or harm any land-use practices and Drainage system as it is an industrial process designed with technical requirements and following the specified norms of local pollution control board.

For Scope-2

Land-use refers to the way in which the land has been used by humans and their habitat, usually with accent on the functional role of land for economic activities. It is the intended employment of management strategy placed on the land-cover type by human agents, and/or managers. Land-cover refers to the physical characteristics of earth's surface, captured in the distribution of vegetation, water, soil and other physical features of the land, including those created solely by human activities; for example - settlements. The land use and land cover are complex and largely continuous pattern and in order to understand its complexity, it is necessary to characterize them.

Land use land cover of the study area is derived through interpretation of satellite remote sensing image in digital environment using remote and GIS software Erdas Imagine (version 8.5). Satellite Image of the study area downloaded from U.S. Geological Survey web site USGS Earth Explorer (www.earthexplorer.usgs.gov) and a land use map was prepared. The satellite remote sensing, with its synoptic view and repetitively, is very helpful in order to cover large areas within a short time to characterize land use & land cover qualitatively and quantitatively.

Results and Discussion

The date of acquisition of the satellite image for the study area is 19th October 2021 with the path and row of 145 and 49, respectively.

Drainage pattern of the study area was developed using Survey of India (SOI) toposheet of the numbers D43E14. Toposheets were downloaded from Online maps Portal of Survey of India website (https://onlinemaps.surveyofindia.gov.in/). The drainage pattern thus developed clearly showed that most of the drainage flows towards **north eastern region** joining Tungabhadra River flowing towards the northern eastern direction of the study area.

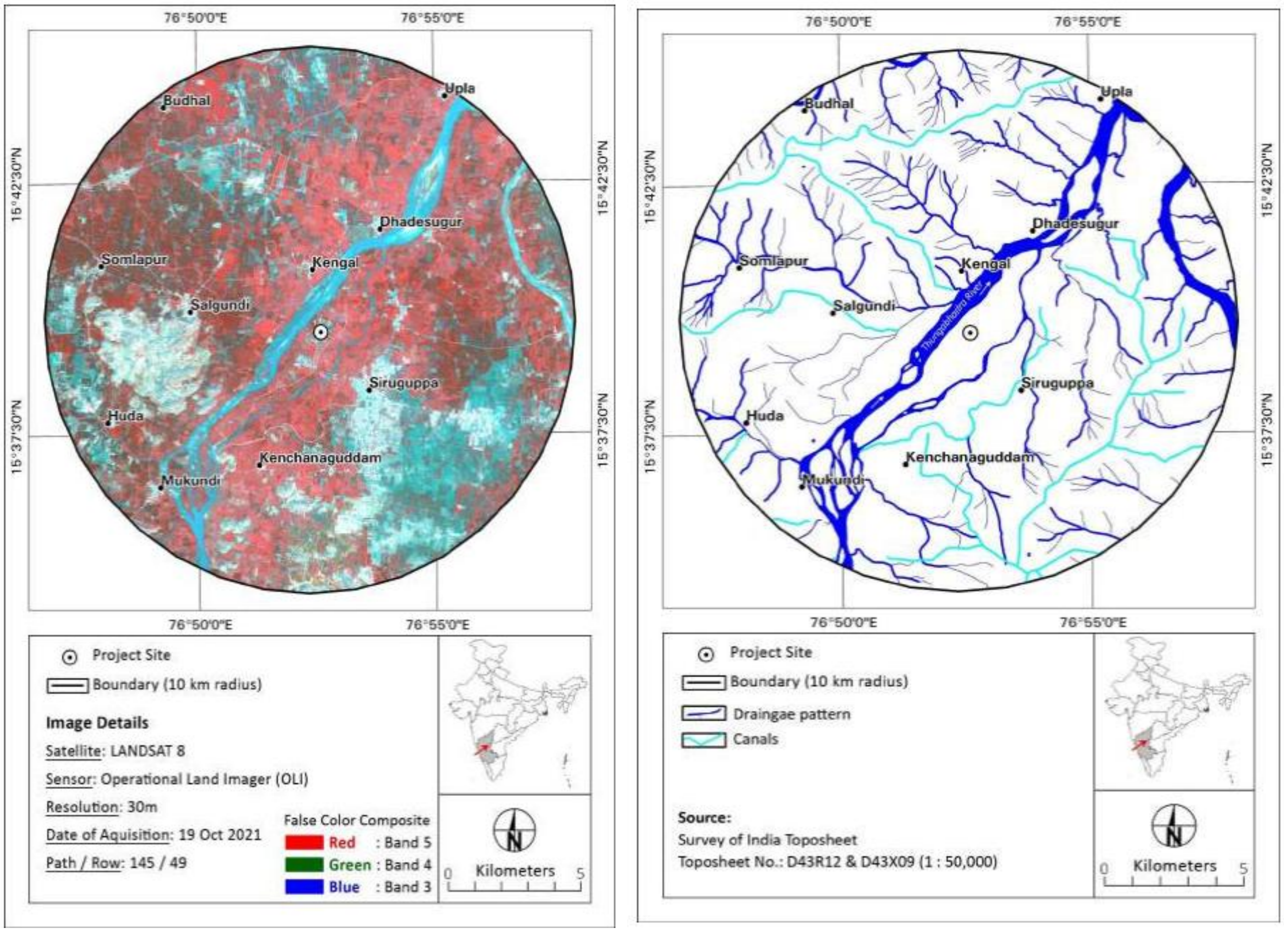


Figure 04: Satellite Image and Drainage pattern of the project site in 10 km radius

The Digital Elevation Model (DEM) for the study area indicates that, the difference in the elevation of the study area is just 129 m above Mean Sea Level only (from 532 – 661 m, Fig. 05) indicating comparatively flat terrain.

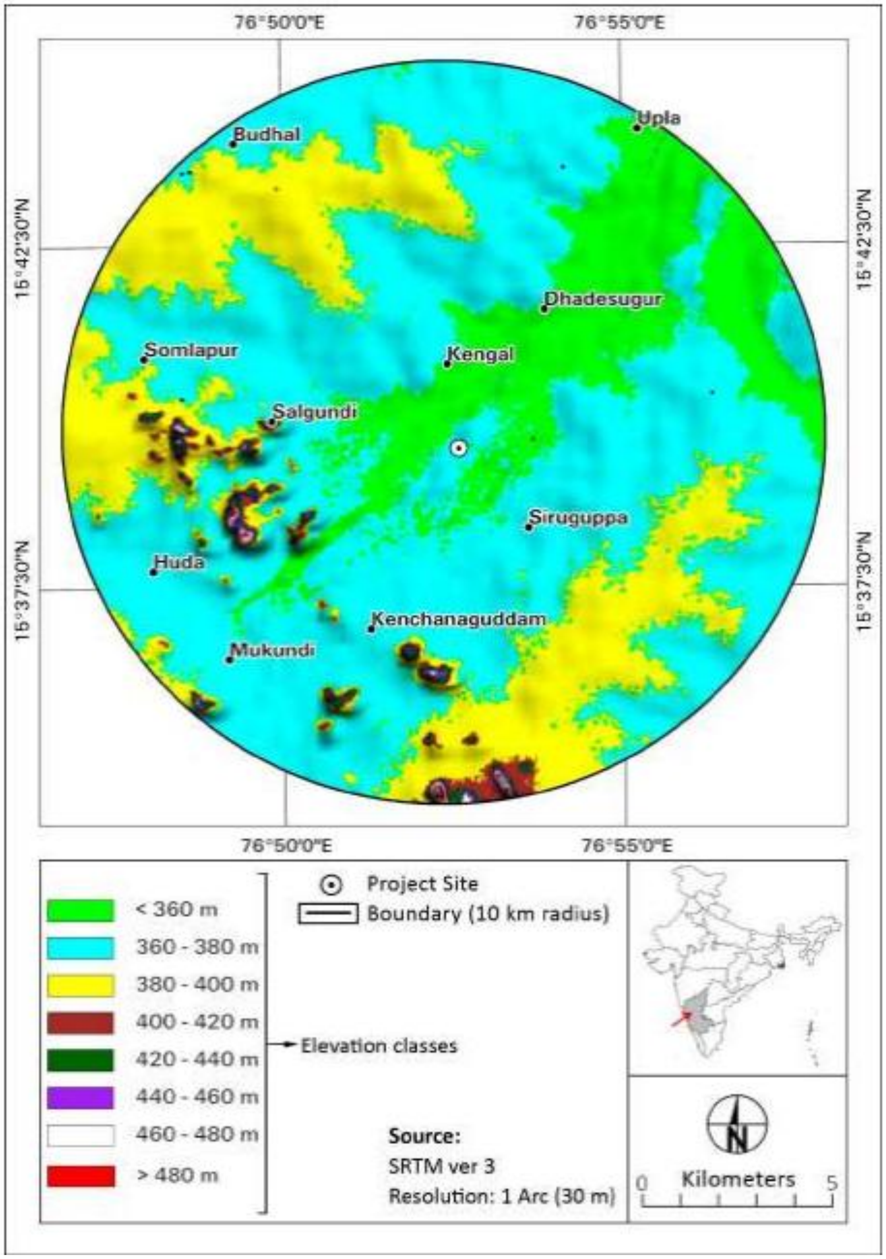


Figure 05: Digital Elevation Model of the project site in 10 km radius

Five different land use / land cover classes have been identified in the study area and the image was classified accordingly. Table 02 shows the information about the extent of identified land use / land cover classes thus derived from the satellite image in the study area and represented below in Figure 06.

Table 02: Land use / land cover classes in 10 km area around the project site

S. No.	Land use / Land cover classes	Area (ha)	Area (%)
1	Agriculture	19,170.38 61.38	61.38
2	Agriculture Harvest	5,462.55 17.49	17.49
3	Open / Barren land	4,142.63 13.26	13.26
4	Built up	790.09 2.53	2.53

5	Water body	1,666.31 5.34	5.34

There are **no reserved forests** in the study area. The project site is almost on the banks of the Tungabhadra River, which flows in the centre of the study area towards the north eastern direction from the project site. This river is breeding ground for many birds. The closeness of the river to the project site is of ecological concern. Utmost care need to be taken to avoid any discharge from the facility (even slightest discharge will affect the flora and the faun of the river). Precautions need to be taken to avoid any possible impacts of the leaching from the industry wastes (both solid and liquid and organic as well as inorganic) generates in the premises of the industry to the river. Here proper care must be taken to ensure no escape of any possible leakage of chemicals to the stream.

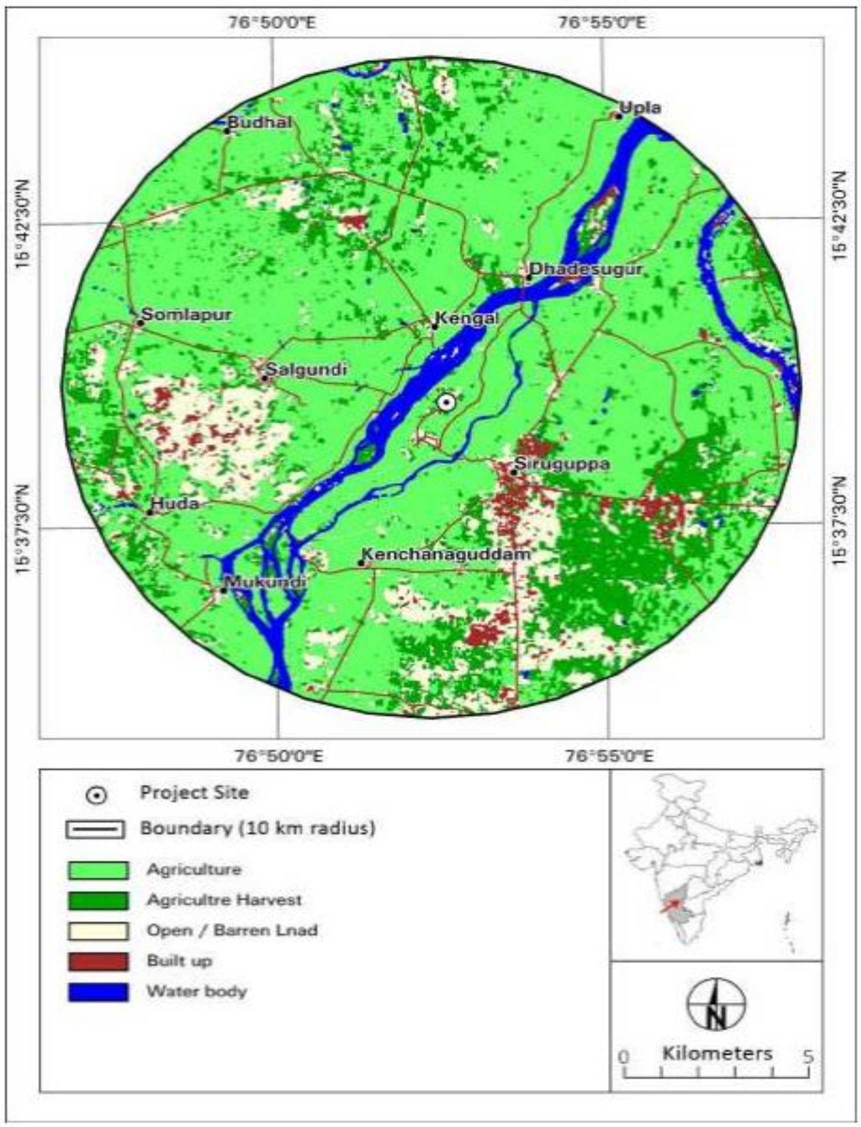


Figure 06: Land use / Land cover map of the project site in 10 km radius

A.4. Climate

For Scope-5

Not Applicable, since this project does not include any activity that is influenced by the climatic conditions of the area as it treats and reuses only the wastewater from the distillery and sugar plant without letting the water exposed to any climatic condition.

For Scope-2

Climate

The climate is a dominant factor in determining the relative abundance or scarcity of water in an area. The available secondary sources of information have been utilized for analyzing the climate characters of the study area. Climatologically the area experiences a **tropical sub-humid type of climate** with moderate summer and good seasonal rainfall. The southwest monsoon sets in the second week of June and lasts till September end. October and November receive rainfall from the NE monsoon. The Winter season with cool and fine weather prevails from December to February, followed by the summer season up to early June.

GMAO generates and distributes some products that either make extensive use of NASA's satellite observations, provide support to satellite missions and field campaigns, or help with the planning for new missions. These products also support researchers funded by NASA and others. This provides quick links to these products: more details are found on the thematic pages.

Long-term, model-based analyses of multiple datasets using a fixed assimilation system are a major focus in the GMAO. Building on the success of the atmospheric reanalyses conducted with GEOS, current research and development activities are directed at producing a major Earth System Reanalysis, including atmosphere, land, ocean, and ice.

The Modern-Era Retrospective analysis for Research and Applications, Version 2 (MERRA- 2) provides data beginning in 1980 and runs a few weeks behind real-time. Alongside the meteorological data assimilation using a modern satellite database, MERRA-2 includes an interactive analysis of aerosols that feed back into the circulation, uses NASA's observations of stratospheric ozone and temperature (when available), and takes steps towards representing cryogenic processes**. Based on the MERRA-2, the Temperature and Relative Humidity figures were obtained for the Plant location and presented below.**

Period	2013	2014	2015	2016	2017	2018	2019	2020	2021	Grand Total
Jan	21.47	20.86	22.25	22.02	21.99	20.21	23.83	22.06	20.81	**21.72**
Feb	22.24	22.72	22.68	22.08	22.39	21.35	22.49	22.20	22.07	**22.25**
Mar	25.00	25.17	24.14	24.77	24.19	23.46	26.00	24.61	23.27	**24.51**
Apr	28.11	27.15	26.93	28.57	27.28	27.02	29.48	28.55	27.27	**27.82**
May	30.17	30.09	27.44	31.68	29.85	29.61	30.08	28.71	28.44	**29.56**
Jun	27.12	26.95	25.97	27.89	27.09	26.43	28.12	28.70	26.05	**27.15**
Jul	23.58	25.35	24.52	24.33	24.70	24.18	26.13	24.87	24.12	**24.64**
Aug	22.74	23.74	24.94	24.01	24.87	24.18	24.73	23.62	23.67	**24.06**
Sep	23.30	23.34	24.93	24.26	24.21	24.01	23.67	23.76	23.84	**23.92**
Oct	22.91	23.54	24.86	24.25	23.48	24.93	23.59	22.89	23.75	**23.80**
Nov	22.92	22.88	24.26	24.87	22.74	23.63	23.06	22.79	23.41	**23.40**
Dec	22.01	21.46	21.96	24.63	21.75	23.46	22.53	21.85	21.85	**22.39**
Grand Total	**24.30**	**24.44**	**24.57**	**25.28**	**24.55**	**24.37**	**25.31**	**24.55**	**24.05**	**24.60**

Table 03: Average Monthly Temperatures of NSL Sugars Tungabhadra Unit

During the winter season from December to February, low temperatures are recorded from a minimum of 20.21°C to a maximum of 23.83°C with an average of 21.72°C. The temperature begins to rise from March and go high during May with a minimum of 27.44°C to a maximum of 29.61°C with an average of 29.99° C. Occasional Thunder storms with breeze bring welcome relief in afternoons. With the onset of the monsoon in early June, the average temperatures begin to drop and minimum being 22.91°C in September and the maximum being 28.12°C in June. However, the average temperatures of the area during the monsoon period vary from 23.92°C to 27.15°C.

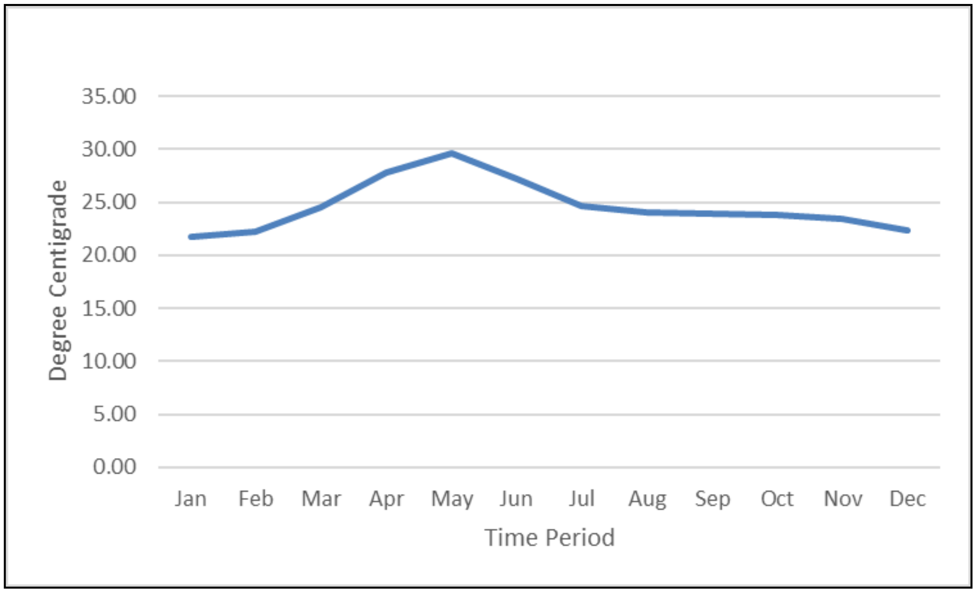


Figure 07: Variation of Temperatures of an area in and around NSL Sugars Tungabhadra Unit

Relative humidity, in general, is moderate to high throughout the year in the area. Humidity is found to be lowest during April, ranging from 31% to 41% with an average of 37.1%. Humidity starts rising in May being between 38% to 47% and continues rising and becoming appreciable up to December, varying between 58% to 82% with an average figure of 63% to 76%.

Over all variation, during the year the humidity varies from a minimum of 37% to a maximum of 76% indicating that the humidity variations are appreciable. Skies are generally heavily clouded during the southwest monsoon, moderately during a post-monsoon period and clear or lightly clouded during the rest of the period. Variation of relative humidity is presented is given in Table 04 and Figure 08.

Period	2013	2014	2015	2016	2017	2018	2019	2020	2021	Grand Total
Jan	62	62	64	70	53	75	53	75	77	**66**
Feb	50	51	49	53	48	58	40	49	69	**52**
Mar	43	42	36	40	33	43	42	39	48	**41**
Apr	38	34	41	38	42	39	31	38	34	**37**
May	41	36	55	36	42	44	38	45	47	**43**
Jun	62	64	68	58	65	68	59	55	69	**63**
Jul	76	72	73	73	70	73	67	71	75	**72**
Aug	79	76	69	74	68	73	69	79	79	**74**
Sep	79	78	69	72	75	71	75	79	76	**75**
Oct	82	78	70	68	82	68	79	82	77	**76**
Nov	83	79	66	57	84	69	84	82	83	**76**

Dec	76	70	81	43	82	63	82	82	87	**74**
Grand Total	**64**	**62**	**62**	**57**	**62**	**62**	**60**	**65**	**69**	**62**

Table 04: Average Monthly Relative Humidity of NSL Sugars Tungabhadra Unit

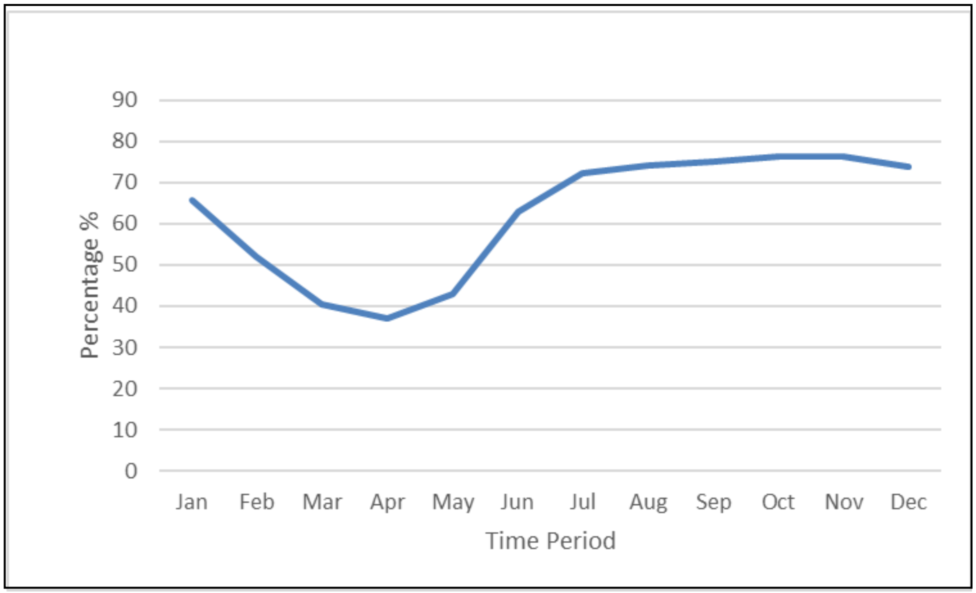


Figure 08: Variation of Monthly Humidity of area in and around NSL Sugars Tungabhadra Unit

A.5. Rainfall

For Scope-5

Not Applicable, since this project does not involve any usage of rainfall data.

For Scope-2

(a) Average annual: 527.4 mm

(b) No. of Rainy days: 75

(c) Temperature: The average temperatures of the area during the monsoon period vary from 23°C to 28°C.

The project activity area gets the benefit of both South West as well as North East monsoon rainfall. The NSL Plant is collecting the daily rainfall data from a **rain gauge station** established in their Plant premises in 2013 and continuing to monitor. The average rainfall data from 2014 to 2025 (12 Years) of the station is 523.17 mm. The maximum rainfall of 734 mm was received in 2016 and a minimum of 232 mm during the year 2019.

The monthly rainfall data of NSL Sugars from 2014 to 2025 is presented in Table 05 below:

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Annual Rainfall (mm)
2014	0	0	0	42	3	77	83	90	164	102	1	0	562
2015	0	0	0	7	85	51	52	158	142	65	4	24	588
2016	0	1	11	8	156	14	56	73	343	56	4	12	734
2017	0	0	1	64	50	197	124	31	87	4	0	2	560
2018	4	0	0	5	13	90	45	101	58	130	0	2	448
2019	0	0	0	11	45	5	20	30	88	33	0	0	232
2020	0	0	0	7	8	26	58	38	264	87	0	16	504
2021	0	0	0	1	4	101	150	62	101	21	1	0	441
2022	0	0	47	9	5	88	82	203	136	114	0	26	710
2023	0	0	6	10	53	16	109	32	112	0	46	0	384
2024	0	0	3	11	41	112	26	117	33	68	7	0	418
2025	0	0	0	1	96	54	70	172	201	86	17	0	697
Average Rainfall (mm)	**0.33**	**0.08**	**5.67**	**14.67**	**46.58**	**69.25**	**72.92**	**92.25**	**144.08**	**63.83**	**6.67**	**6.83**	**523.17**

Table 05: Monthly Rainfall Data of NSL Sugars Tungabhadra Unit (2014 to 2025)

In general, the rainfall for 12 years from 2014 to 2025 is showing a rising trend. The year 2017 and successive three years followed by 2021 being good rainfall years, general rainfall is showing a rising trend.

A.6. Ground Water

For Scope-5

Not Applicable, since this scope of project does not include any activity that uses the ground water.

For Scope-2

The Ground Water Resources Estimated by CGWB (Following GEC Methodology 2015) for the year 2020, the Annual Ground-Water Resources of the Siruguppa Taluk is 8154 Ham. The Total Ground Water Extraction for irrigation, domestic and other uses is 4493 Ham. The net ground water available for future use is 3661 Ham. The Stage of Ground Water Extraction of the entire Siruguppa Taluk is 55% and classified as Safe Category indicating that there is a scope for further ground water extraction in the area. However, the mandatory guidelines like rainwater harvesting and artificial recharge issued by Karnataka Ground Water Authority need to be strictly implemented in the area, so that ground water sustainability is maintained.

Hydrogeology

Hydrogeological conditions of the area are also among important factors in planning rainwater harvesting aquifer recharge schemes. The recharged water moves below the soil zone in moisture fronts through the zone of aeration. The unsaturated flow is governed by the permeability of the zone of aeration, which in turn varies with the moisture content of the front. Usually, in the case of consolidated rock formations, the subsoil zone passes into weathered strata, which, in turn, passes into un-weathered but fractured rock. The hydrogeological properties of the weathered strata are generally much better as compared to the parent rock due to higher porosity and permeability imparted by weathering. The nature of the soil, subsoil, weathered mantle and presence of hard pans or impermeable layers govern the process of recharge into the unconfined aquifer.

The saturation and movement of ground water within unconfined and all deeper semi- confined and confined aquifers are governed by the Storativity and Hydraulic Conductivity of the aquifer material. Aquifers best suited for artificial recharge are those, which absorb large quantities of water and release them whenever required. Detailed geology and hydrogeological study besides the regional picture of hydrogeological set up available from previous studies is therefore imperative to know precisely the promising hydrogeological units for recharge and correctly decide on the location and type of structures to be constructed either for storage of rainwater harvested or for recharge to the ground water.

Regional Hydrogeology

Ground water occurs in all the formations. However, nature and occurrence depend on various factors like rainfall, topography, land forms and geology of the area. From the ground water point of view, the geological formations occurring in the area may be classified into two categories as **Hard Rock** (crystalline Rocks) and unconsolidated sediments or **soft rocks.**

Crystalline rocks consisting of granite gneisses, schists and Charnockites are grouped into hard rocks, while **unconsolidated formations** especially the alluvium along the rivers and stream courses are the soft rocks.

The hard crystalline rocks lack primary porosity. The occurrence and movement of ground water are usually limited to secondary porosity developed through weathering and fracturing of the rock formation. As a result, groundwater prospects in hard rocks are limited and depends upon the degree of the intensity and depth of fracturing and weathering. The intensity and depth of weathering and fracturing vary widely from place to place. The hard rock aquifers are anisotropic, and non-homogenous, as such occurrence and movement of ground water vary widely within a short distance. Ground water occurs under water table conditions in weathered residuum and semiconfined to confined conditions in deeper fractured zones. Generally, most of the area is underlain by hard rock aquifers.

Shallow zone aquifers occurring within the depth of 25 m below ground level are constituted of weathered and fractured granite gneisses, granites and schist. Ground water occurs in the open spaces of weathered and fractured formations under water table conditions. The groundwater of this zone is utilized through structures like dug wells, dug cum borewells and shallow borewells.

The moderately deep aquifers constituted of weathered and fractured granite gneisses, granites and schist. Ground water occurs in the open spaces of weathered and fractured formations under semi-confined conditions.

The aquifers occurring within the depth of 200 m below ground levels are grouped as deep aquifer zones. The aquifers of this category constituted of fractured and jointed granite gneisses, granites and schist. Ground water occurs in the open spaces of fractured and jointed formations under semi-confined to confined conditions.

The site under investigation is regionally the area underlain by geological formations ranging in age from Archaean to Recent period mostly consisting of Granitic Gneiss, and Dharwar Schists to recent alluvium. Various intrusive rocks like quartz veins and dykes have traverse these formations later. The granite gneisses are occurring extensively in the area, whereas schists occur as north-south trending in a linear direction as patches underlain by schistose rock formations.

The Plant and its surrounding areas area are underlain by the rock of Dharwar schistose formation extending in a north-south narrow and linear fashion in the central part of the district along a thrust zone are extending to the present study area. The schists are sandwiched between the gneisses on both sides. The schists are generally composed of predominantly a volcano-sedimentary sequence intruded by meta gabbro dykes and younger quartz veins. The soils are red in colour, and gravelly in nature followed by the weathered schist rock.

Weathering and fracturing of these rocks form the aquifer in the area. The weathered portion in schistose rocks is relatively less porous than other rock formations like Gneisses. The nature of the soil, subsoil, weathered mantle and presence of hard pans or impermeable layers govern the process of recharge into the unconfined aquifer. The saturation and movement of ground water within unconfined and all deeper semi-confined and confined aquifers are governed by the **Storativity and Hydraulic Conductivity** of the aquifer material. As per the exploratory well drilling carried out by CGWB at Kokkera Bellur, Siruguppa, Kestur and B Hosur, to a depth of 140 to 200 m bgl which are nearby areas to the Plant, revealed that the average depth of weathering varies from 15.0 to 24.0 m bgl. The yield of the bore wells ranged from a minimum of 0.046 lps to 4.3 lps, however, the overall average yields of the bore wells remained between 1.0 lps to 2.0 lps.

There are **two aquifer** systems in the area.

- The First aquifer weathered aquifer down to a depth of 24 m bgl. The ground water in this zone occurs under phreatic conditions.
- The first aquifer is followed by the second aquifer constituting fracture zones from 30 mbgl to a maximum of 188 m bgl. The occurrence of ground water in this zone is generally found to be under semi-confined to confined conditions.

The transmissivity of the aquifer was found to be 2.28 m^3/day to 195 m^3/day with an average of 20 to 30 m^3/day.

The storage co-efficient values of the granite gneiss and schist aquifers existing in the area varied from 6.98×10^{-4} to 2.42×10^{-4} indicating that the deeper aquifers are under confined condition. The aquifer parameters like transmissivity storage co-efficient and yield of the bore wells indicate that the area has poor to moderate yielding aquifer to a depth of 200 m bgl.

The average depth to ground water levels in the first aquifer, generally, in open wells during the pre-monsoon period is varying from 1.5 to 11.6 m bgl, whereas during the post-monsoon period it is ranging from 1.07 to 5.0m m bgl. Similarly in the second aquifer, i.e fractured aquifer, generally in bore wells, the average ground water levels remain at 5.51 m bgl during the pre- monsoon period and 4.66 mbgl during the post-monsoon period. In both the aquifers the general ground water levels are found to be moderately deep.

There are **six** bore wells in the plant premises. The depth of the bore wells is varying from 60 to 70 m bgl. The wells located at Dormitory and ETP are not being pumped, whereas the wells located at Coal, Bagasse, 9 acres area are being operated 2 to 3 hours in a day. The well located at R&D-3 is generally pumped for about 8 hours a day and a major part of the water requirements to the residential area is met through this bore well. These wells are moderate yielding with the discharges ranging from 1.0 to 2.5 lps (5 to 8 m^3/hour). The overall ground water quality is good in the area.

The Central Ground Water Board is monitoring an observation well on regular basis to monitor the ground water levels and to assess the general ground water regime of the area. One such Monitoring well is located in Siruguppa Village which is very nearest to the present area under consideration. The data is available from May 2000 to October 2020. The shallowest ground water levels with an average of 1.5 m bgl were observed during November, the post-monsoon period after rainfall. Similarly, the average deep water level of 2.70 m bgl was found to be during May in the summer period.

Overall, in **Siruguppa station**, the depth to water levels are relatively shallow, when compared to the surrounding areas. The long-term water level data indicates that the water levels are generally rising at the rate of 0.07 m/year which is very meagre over 10 year’s period.

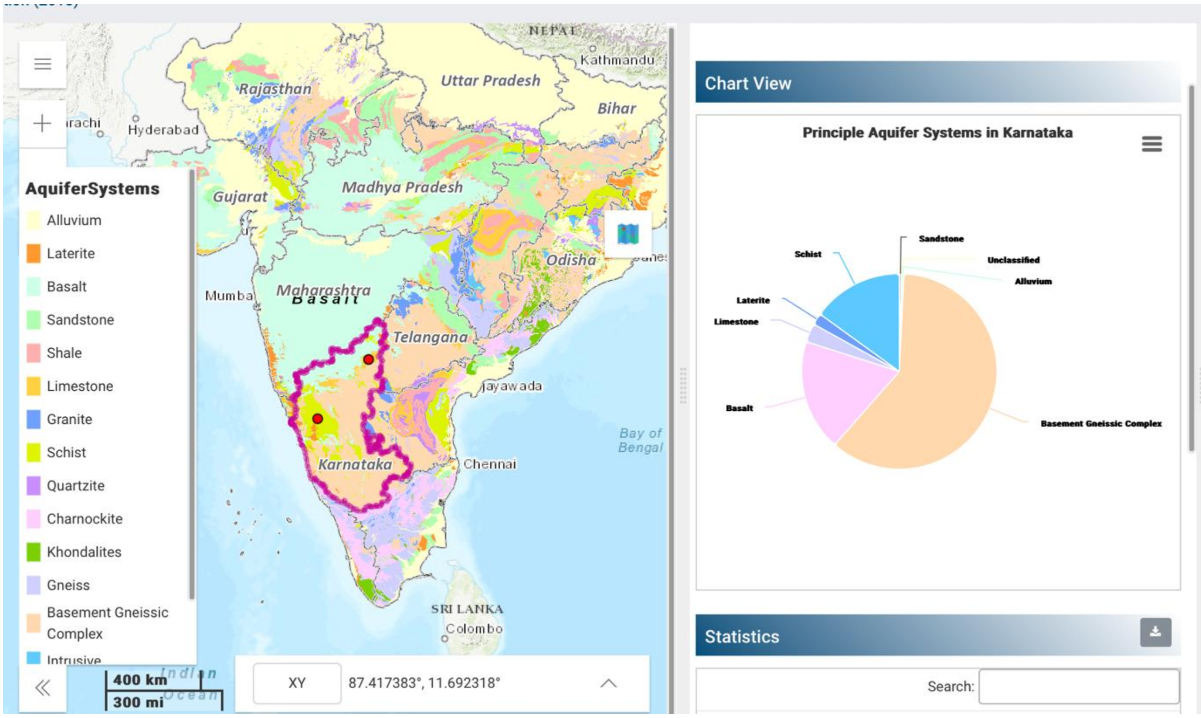


Figure 09: The Aquifer system available according to the India Water Resources Information System[3] in the plant site state- Karnataka

Statistics

Search:

Principle Aquifer	Aquifer System	Aquifer Type	Thickness (m)	DTW (Decadal Avg.) (mgbl)
Alluvium	704.44	Unconfined	20-60	20-60
Basalt	34899.50	Unconfined	10-45	10-45
Basement Gneissic Complex	114967.88	Unconfined Semi-confined to Confine	Up to 60	Up to 60
Laterite	3627.27	Unconfined	5 -15	5 -15
Limestone	5873.49	Unconfined, Semi Confined to Confin	40-150	40-150

[3] https://indiawris.gov.in/wris/#/Aquifer

Sandstone	721.04	Unconfined	60-80	60-80
Schist	28464.07	Unconfined-Semi-confined	35	35
Unclassified	38.29			

Showing 6 to 8 of 8 entries Previous 1 2 Next

Figure 10: Showing the types of Aquifer system and their technical specification available according to the India Water Resources Information System in the plant site state- Karnataka

A.7. Alternate methods

W.r.t Scope-5

a) Common Practices for Wastewater Treatment in Sugar and Distillery Industries:
- Anaerobic Digestion: Used to treat high-strength effluent like spent wash, reducing COD/BOD and generating biogas.
- Multiple Effect Evaporation (MEE): Concentrates spent wash to reduce volume, preparing it for further treatment.
- Bio-composting: Combines treated spent wash with press mud to produce organic fertilizer.
- Zero Liquid Discharge (ZLD): Integrates MEE, crystallization, and reverse osmosis to recover water and prevent discharge.
- Irrigation Use: Treated effluent is diluted and applied in agriculture under controlled conditions.
- Incineration: Concentrated spent wash is burned for energy recovery in distillery operations.

b) Common Practices for Condensate water in Sugar and Distillery Industries:
- It is generally used in the process back after treatment.
- In some distilleries especially molasses-based units, condensate can be reused for diluting molasses in the fermentation section.

These methods align with regulatory norms and support resource recovery and sustainability.

Industries face challenges such as high capital and operational costs, land scarcity for methods like bio-composting, and skilled workforce shortages. Managing spent wash with high COD/BOD, odor control, seasonal effluent variations, and strict regulatory compliance adds complexity. Additionally, handling byproducts and ensuring sustainable water reuse remain significant concerns. Hence implementing these methods in industry improves the water security.

W.r.t Scope-2

As per the study and data available of the NSL project it is clear that the project owner is trying to collect and conserve the rainwater for reuse in the plant activity to avoid any runoff and wastage of water. The hydrological structures and conservation measures are developed in such a way that the runoff generated within the plant will be collected and stored to meet a part of the water requirement of the Plant.
As it was determined from each zone and land use hydrological characteristics, the runoff is the major component which is being affected by various factors. The following are the factors which are affecting different hydrological characteristics:

Storm Characteristics	Nature of Storm
	Seasonal Storms
	Intensity
	Duration
	Areal extent (Distribution)
	Antecedent Precipitation
	The direction of Storm Movement
Meteorological Characteristics	Temperature
	Humidity
	Wind velocity
	Pressure Variation
	Size
	Shape
	Slope (Average land slope & Stream slope)
	Altitude (Elevation)
Catchment Characteristics	Topography
	Geology
	Land use/vegetation
	Orientation
	Type of drainage network
	Proximity to Ocean and mountains
	Ranges
	Depressions
	Pits and Ponds
Storage Characteristics	Steams and channels
	Upstream tanks/ponds
	Flood plains and swamps
	Ground water storage in pervious deposits

Based on the above characteristics it is found that in general, low-intensity storms over longer spells contribute to the ground water storage and produces relative less runoff. On the other hand, high-intensity storm or smaller catchment areas increases the runoff since the losses like infiltration and evaporation are less. However, if there is a succession of storms the runoff will increase due to the initial wetness of the soil, and antecedent rainfall and most of the area is well developed into buildings for Plant activities and paved ones. As usual, the rain during the summer will produce less runoff and during the winter will produce more runoff.

The development of these catchment areas and measures to prevent run-off will save water and result is less runoff in the area.

A.8. Design Specifications

W.r.t Scope-5

A) Sugar Effluent Treatment Plant (ETP) is designed to treat wastewater generated from leakages and spillage of juice and syrup and molasses in different sections of the industry. Leakage occurs at pipe joints and pump glands. The periodical washing of floor also contributes pollution load in the wastewater. Wastewater is also generated due to the cleaning of equipment such as mills, evaporators, pans, juice heaters, etc. The effluent thus generated is having COD and BOD that needs to be treated before using it for agricultural purposes.

ETP is having following 9 steps of treatment processes:

1. Screen Chamber, Oil & grease trap
2. Equalization Tank
3. Monthly washings tank
4. Neutralization tank
5. Tube settler tank & Feed tank
6. Anaerobic Digester
7. Extended Aeration Tank
8. Secondary Clarifier
9. Treated Effluent Storage tank

Details of treatment unit:

S. No.	Name of the unit	Dimensions	Volume, m^3
1	Bar Screen chamber	2.0m X 1.0 m X 0.7m SWD	1.4
2	Oil & grease removal	2.0 m X 8.5m X 1.2m SWD	20.4
3	Equalization Tank	12.9 mX12.9m X3.0 SWD	500.0
4	Monthly Washings tank	15.15m X30.0m X 3.0m SWD	1500
5	Neutralization tank	3.75m X 3.75m X 3.0m SWD	42.2
6	Tube settler tank	3.5m X 3.9m X2.0 m SWD	41.0
7	Feed Tank	3.9m X 3.9mX 2.7m SWD	41.7
8	Anaerobic Digester	8.0m Dia X 8.9m SWD.	447.1
9	Extended Aeration tank	11.2m X 22.3m X 3.6m SWD	900.0
10	Secondary Clarifier	10.0m Dia X 2.5m SWD	196.2
11	Treated effluent storage tank	19.0m X 21.9m X 2.1m SWD	1000.0
12	Sludge Drying Beds	5.0m X 5.0m X1.0m SWD	200
13	Sludge holding tank	10.m X3.3m X 2.7m SWD	99.0

Table 06: Major ETP tank list of NSL Sugars Tungabhadra Unit

Physio-Chemical process:

1) Screen Chamber, Oil and grease trap:

Screen chamber has been provided to accommodate the screen. The screen will remove the suspended particles like bagasse, cotton waste, etc. coming from the sugar unit.

BAR SCREEN CHAMBER:

Type : Rectangular

Volume	:	1.4 M^3
Size	:	2000 x 1000 x 700mm SWD
MOC of Tank	:	R.C.C. M20
Screen	:	Manual Removable type
MOC of Screen	:	SS
Quantity	:	1 No.

Oil & Grease trap is provided for removal of free & floating oil and grease, which otherwise would affect the performance of anaerobic biological treatment. The trap is provided with oil removal mechanism. A screen chamber is provided ahead of oil & grease trap for removal of coarse material.

Type	:	Rectangular
Volume	:	14 M^3
Size	:	2000 x 8500 x 1200mm
MOC of Tank	:	R.C.C. M20
Mechanism	:	Belt Type
Quantity	:	1 No.

2) Equalization Tank:

An Equalization Tank is provided for dampening the fluctuations in wastewater characteristics and quantity. From this tank the effluent will be pumped to neutralization tank.

Type	:	Square
Volume	:	549 M^3
Size	:	12900 x 12900 x 3300mm
MOC of Tank	:	R.C.C. M20
Quantity	:	1 No.
Pump Capacity	:	42 M^3/Hr
Type of Pump	:	Self-priming, non-clog, semi open impeller
Quantity	:	2 Nos (1W+1Sb)
Mixing	:	By Air

Operating parameters- Feed pH should be maintained between 6.8 to 7.2 by adding Lime

3) Surge tank/Monthly washings tank:

Evaporators, Juice Heaters, Pans etc. need to be cleaned once in a month for removal of scale.
Caustic soda, Soda Ash and Hydraulic Acid are used to remove scale. Because of high pH and pollution load this wastewater is collected in a separate tank called monthly wash tank and on an average 25-30 m^3/day of this wastewater is drawn from the monthly wash tank and added to the equalization tank.

Type	:	Rectangular
Volume	:	1500 M^3
Size	:	15150 x 30000 x 3300mm
MOC of Tank	:	R.C.C. M20
Quantity	:	1 No.
Pump Capacity	:	10 M^3/Hr
Type of Pump	:	Self-priming, non-clog, semi open impeller

Quantity	:	2 Nos (1W+1Sb)

4) Neutralization tank:

In this tank lime will be added to the effluent to increase the pH of the effluent.

Type	:	Square
Volume	:	42.2M^3
Size	:	3750 x 3750 x 3000 mm
MOC of Tank	:	R.C.C. M20
Quantity	:	1 No.

5) Tube settler tank:
In tube settlers the settling velocity of the suspended particles can be increased. The settling of suspended solids is more efficiently carried out:

Type	:	Rectangular
Surface Area	:	12.25 M^3
Size	:	3900 x 3500 x2000
Volume provided	:	27M^3
MOC of Tank	:	R.C.C. M20
Quantity	:	1 No.

Operation:
The bottom sludge should be drained on daily basis

Feed tank:

The neutralized effluent from the feed tank will be fed to anaerobic digester through pumping.

Type	:	Square
Surface Area	:	42 M^3
Size	:	3930 x 3930 x2700
MOC of Tank	:	R.C.C. M20
Quantity	:	1 No.

6) Anaerobic Digester:

In this type of treatment effluent is treated anaerobically in air tight RCC tank. The reactor consists of three zones viz. Influent distribution zone, Reaction zone, Gas solid liquid separation zone.

Influent Distribution zone: The raw wastewater enters into the digester at the bottom through influent distribution zone. A sophisticatedly designed piping network is provided for uniform distribution of the effluent in the tank. The effluent then travels upward in the reactor.

Reaction zone: In the reaction zone the anaerobic bacteria are maintained on the surface of the PVC media. The organic matter in the wastewater comes in contact with the bacterial population and is degraded anaerobically to methane rich biogas, the end product of anaerobic digestion. The process of conversion of organic matter into the biogas is a two-stage process. In the first stage the organic matter in the raw effluent is converted in to the volatile acids by acid forming bacteria. In the second stage the acid produced in the first stage are converted in to methane by another group of bacteria i.e. methane formers. In anaerobic digester process both the stages are completed in single reactor. The biogas so produced is bubbled through the effluent and is separated out in the third section i.e. Gas collection zone. The suspended solids are also separated to prevent escape of solids from the reactor.

Gas collection zone: In gas collection zone the biogas generated will be collected and the same will be conveyed to flare stack where it is flared continuously. The treated effluent overflows through a launder and will take to a secondary treatment.

Size of the tank	:	8000 mm dia. x 8900 mm ht.
Tank Vol	:	447 M^3
Gas generated per Kg	:	0.45 NM^3
COD Reduction	:	65 %
Gas Generated	:	0.5 to 0.53 kg per kg of COD reduction
MOC of Tank	:	R.C.C. M 20
Quantity	:	1 No.

Operating parameters:

Feed pH	:	6.8 to 7.5
Digester over flow	:	6.9 to 7.6
VFA to Alkalinity ratio	:	0.3 to 0.6
COD	:	inlet and outlet

7) Extended Aeration tank:

The partially treated effluent from anaerobic digester shall then be subject to activated sludge process for further reduction of organic matter. Aeration tanks are provided for degradation of organic matter through biological process. Microorganism in the controlled environment carries out the biodegradation process. The container i.e. aeration tank of requisite capacity is provided for this purpose. The tank shall be provided with an diffused aeration mechanism to transfer the oxygen from air to tank content for survival of microorganisms. Constant Mixed Liquor Suspended Solids (MLSS) shall be maintained in definite proportions at aeration tank by recycling the bio-sludge trapped by the secondary clarifier. The content in the aeration tank is kept under constant aeration and mixing. 25.00 HP diffused type of aerators shall be provided for transferring the oxygen for metabolism.

Type	:	Rectangular
BOD % Reduction	:	96%
MLSS	:	4500 ppm
Volume	:	900 M^3
Size	:	11200 x 22300x 3600 mm
Air Required	:	858 M3/Hr.*
Energy Required	:	14 KWhr
Aerator Length Required	:	160 meters
Size of each Diffuser	:	2 meters
No. of Diffuser	:	80

Operating parameters:

Feed pH	:	7.6 to 8.2
Settling	:	300 to 500 ml/Litre
MLSS	:	3000 to 4500 mg/l

Colour	:	Brownish
SVI	:	50 to 100

8) Secondary Clarifier:

A secondary Clarifier in the form of circular tank shall be provided for settlement of fully aerated effluent from the extended aeration tank. The tank shall be provided with centrally driven fixed bridge type clarifier mechanism. Part of the settled sludge at the bottom of the settling tank will be pumped to the EAT and part of it will be discharged on sludge drying beds as per operational requirement. This sludge being fully mineralized is suitable for sun drying on sand drying beds.

Type	:	Circular
Total Size	:	10000 mm dia. x 2500 mm SWD
Tank Vol.	:	196.25 M^3
Qty.	:	1 No.

Operating parameters:

pH	:	7.6 to 8.2
Total Suspended Solids:	:	<100 mg/l
COD	:	<250 mg/l
Oil & Grease	:	<10mg/l
Colour & Odor	:	No colour and Odor

Sludge Drying Beds:

In the activated sludge process system the sludge is sufficiently mineralized and does not need any further treatment before dewatering and disposal. Sand filtration drying beds will be provided, where sludge will be dewatered by filtration through sand bed and sun drying of the dewatered sludge is scraped & may be used as manure.

Sludge from Primary Treatment	:	400 Kg dry basis
Area Provided	:	200 M^2
Drying Cycle	:	2 to 3 days

9) Treated Effluent Storage Tank:

This tank is provided to store the treated effluent before pumping to the green belt development area or sugar cane irrigation

List of Equipment:

S. No.	Name of the equipment	Capacity, m^3/hr
1.	Raw effluent transfer pump, Make-Johnson Model: FRE-60-180	Cap-42 m^3/hr
2.	AD feed pump, Make-Johnson Model KGEN	Cap-50 m^3/hr
3.	Bio-sludge recirculation pump, Make: Johnson Model: KGEC	Cap:45 m^3/hr
4.	Treated effluent transfer pump, Make: Kirloskar	Cap:44m3/hr
5.	Air Blower-I (Aeration tank), Make-CVT	Cap:839 m^3/hr Delivery pressure:0.45 Kg/cm^2
6.	Air Blower-II , Make-CVT	Make-CVT, Cap:150 m^3/hr Delivery pressure:0.4 Kg/cm^2

Table 07: Major List of Equipment used in ETP of NSL Sugars Tungabhadra Unit

Final Effluent Quality & Compliance

The treated water from the polishing pond meets environmental discharge norms and can be safely released or reused for agricultural and gardening purposes.

Parameters	UOM	Before Treatment	After Treatment
Quantity	m^3/day	350	350
pH		4.5-5.5	6.5-8.2
Chemical Oxygen Demand	mg/l	5000	<250
Bio-Chemical Oxygen Demand	mg/l	2000	<100

Suspended solids	mg/l	600-800	<100
Oil Grease	mg/l	70	<10

Table 08: Parameters values before and after testing of ETP process at NSL Sugars Tungabhadra Unit

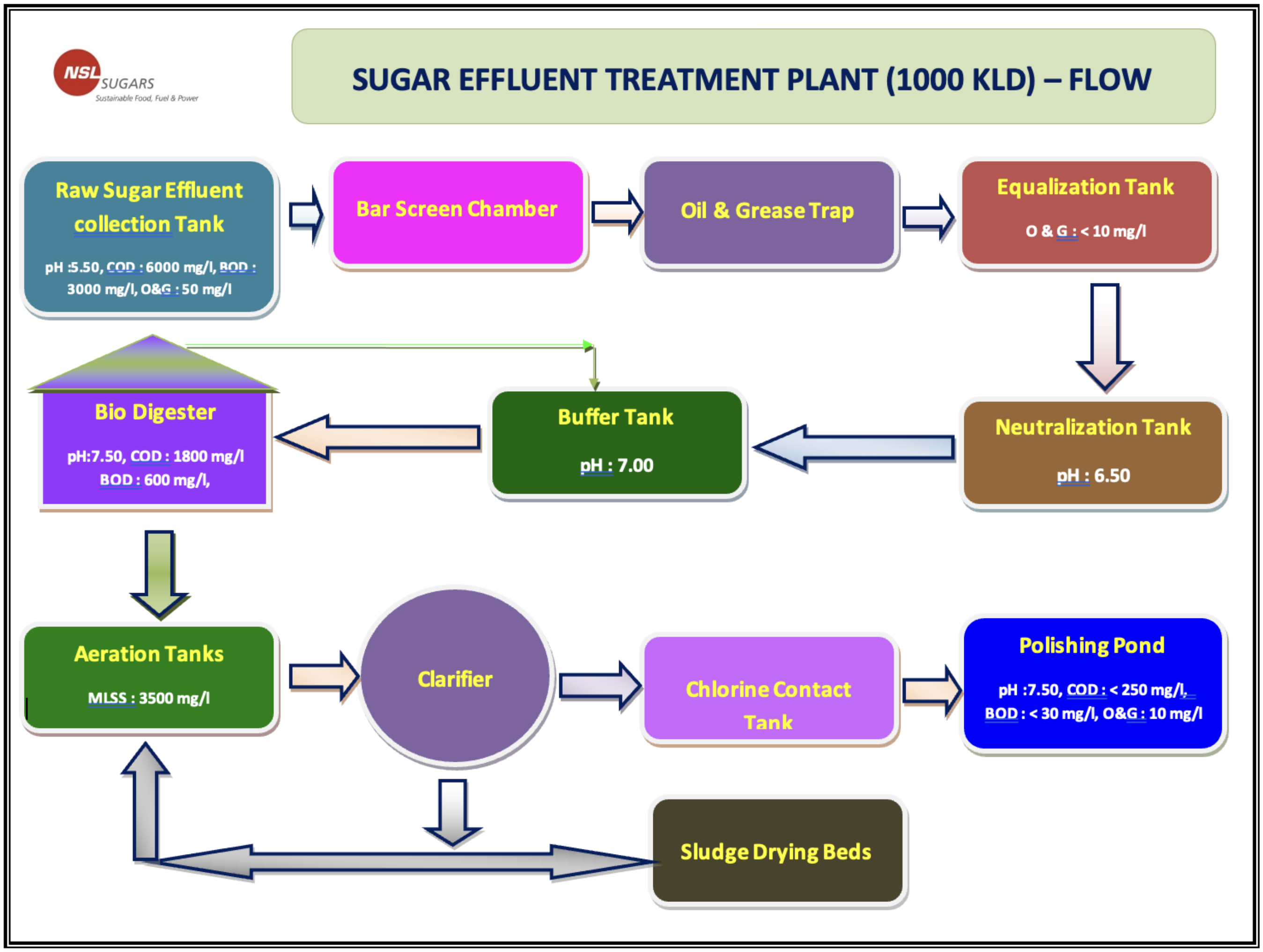


Figure 11: Showing the process flow of STP plant in NSL Sugars Tungabhadra Unit

B) Sugar Condensate Plant: is designed to captures process vapours from evaporators and pans, converting them into purified, reusable water. This drastically reduces fresh water consumption and helps achieve Zero Liquid Discharge (ZLD) when integrated with the distillery.

The major impurity contents of sugar factory condensate are trace organics and more particularly the Ammonical Nitrogen which can prove detrimental for reuse applications.

The treatment scheme for condensate polishing unit involves the utilization of biological mode, specially developed to treat not only trace organics, but basically the Ammonical nitrogen. The treatment scheme therefore comprises of an Aerobic – Anaerobic approach.

Although, the characteristics of condensate are fairly uniform with reference to period, it is desirable to provide some cushion for minor variations in the form of equalization tank. This also serves as a part of composite and complex biological system specially developed as the treatment scheme. The condensate is subjected to aerobic and anaerobic biological modes selectively to cater for and treat the two main impurities. Care is taken to optimize the treatment structures as well as the energy inputs, thereby minimizing the capital as well as working expenditures

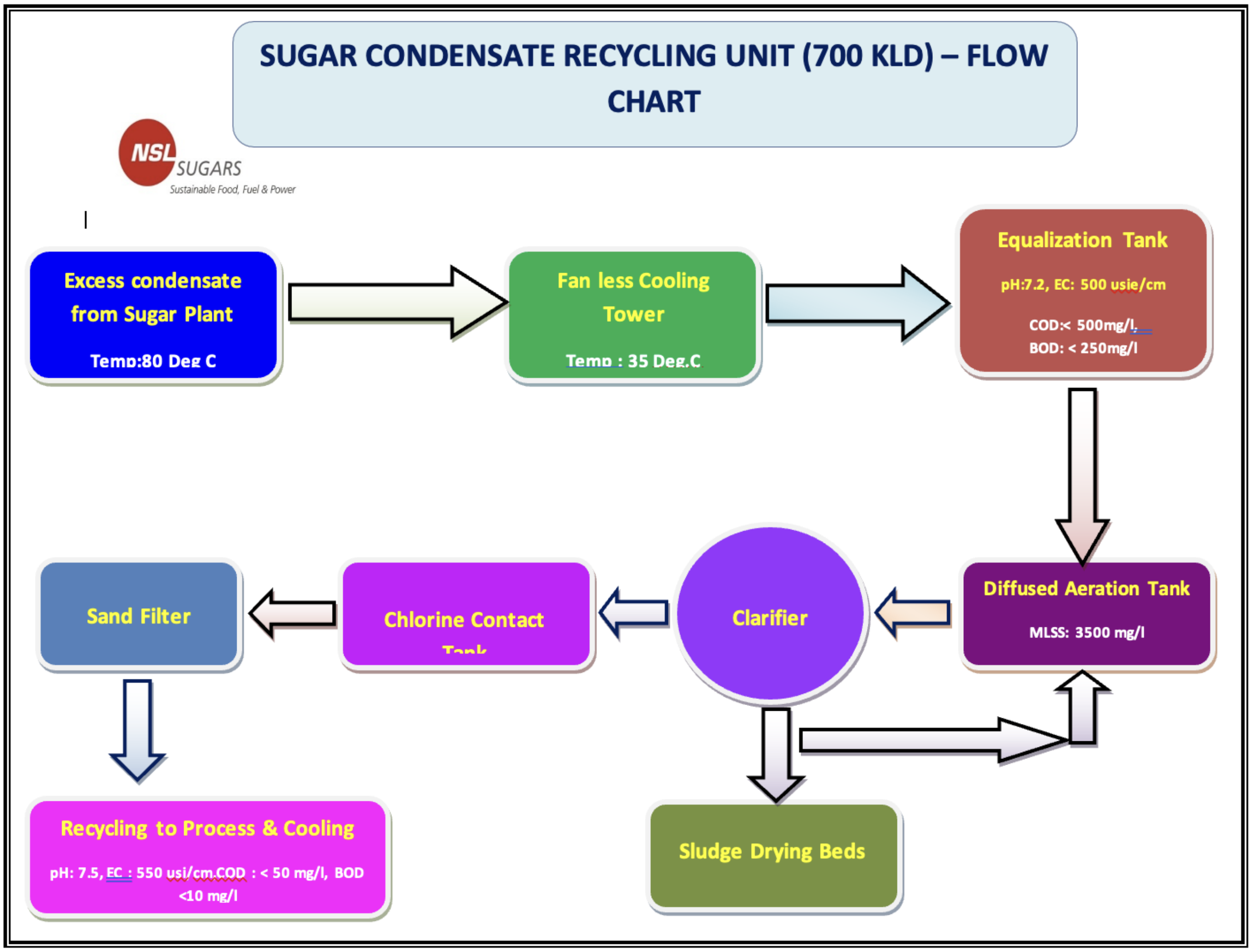
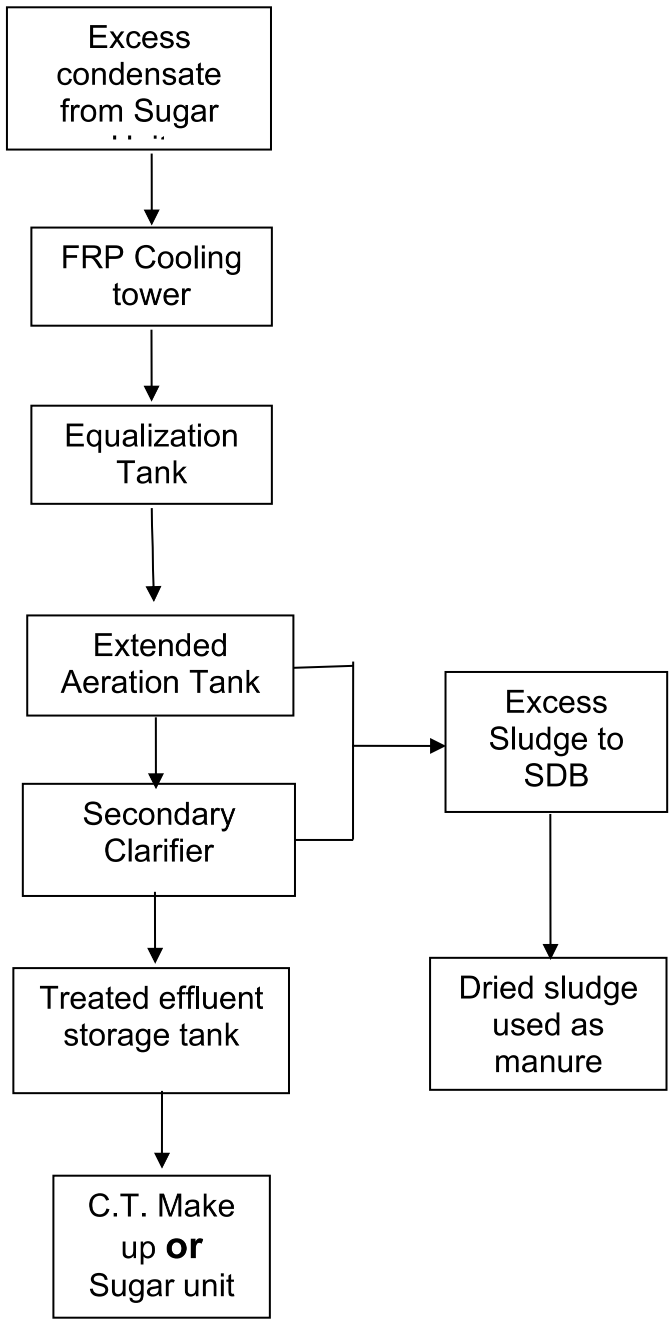


Figure 12: Showing the process flow of Sugar Condensate Recycling plant in NSL Sugars Tungabhadra Unit

SCHEMATIC DIAGRAM OF CONDENSATE POLISHING UNIT



a) Excess Condensate from Sugar Unit:

Condensate generated from various stages of sugar manufacturing, particularly from evaporators and pans, is collected as excess condensate. Although relatively clean, this condensate contains traces of sugars, dissolved organic matter, and volatile compounds, making it unsuitable for direct reuse. Therefore, it is routed to the condensate polishing unit for treatment and recovery.

b) FRP Cooling Tower:

The hot condensate is first passed through an FRP (Fibre Reinforced Plastic) cooling tower to reduce its temperature. Cooling is essential to create favourable conditions for subsequent biological treatment processes and to prevent thermal shock to microorganisms responsible for degrading organic contaminants.

c) Equalization Tank:

The cooled condensate is then transferred to the equalization tank, which acts as a buffer to balance fluctuations in flow rate and pollutant concentration. Continuous mixing in the tank ensures a homogeneous wastewater stream, thereby providing a steady and uniform feed to the downstream treatment units.

d) Extended Aeration Tank:

From the equalization tank, the condensate enters the extended aeration tank where air is supplied continuously through diffusers or aerators. Under aerobic conditions, microorganisms metabolize the dissolved and suspended organic matter present in the condensate, resulting in a significant reduction in BOD and COD. The extended retention time enhances the treatment efficiency and ensures stabilization of the wastewater.

e) Secondary Clarifier:

The mixed liquor from the aeration tank is conveyed to the secondary clarifier, where biological solids settle under gravity. The clarified water overflows from the top and is collected for reuse, while the settled sludge is concentrated at the bottom. A portion of the sludge is recirculated to maintain an adequate microbial population in the aeration tank, and the excess sludge is withdrawn for further handling.

f) Excess Sludge to Sludge Drying Bed:

The excess sludge generated during the biological treatment process is directed to sludge drying beds. In these beds, moisture is removed through evaporation and drainage, resulting in a reduction in sludge volume and facilitating its handling and disposal.

g) Dried Sludge Used as Manure:

After sufficient drying, the stabilized sludge contains organic matter and nutrients that make it suitable for use as manure or soil conditioner. Its beneficial reuse contributes to resource recovery and supports sustainable waste management practices.

h) Treated Effluent Tank:

The clarified water obtained from the secondary clarifier is collected in the treated effluent tank, which serves as a storage and balancing reservoir. This tank ensures a continuous supply of treated water and allows for monitoring of water quality parameters before reuse.

i) Cooling Tower Makeup for Sugar Unit:

Finally, the treated condensate is reused as cooling tower makeup water within the sugar plant. Recycling the treated condensate reduces freshwater consumption, minimizes wastewater discharge, and promotes efficient water management, thereby contributing to the sustainability and resource conservation goals of the sugar mill.

For Scope-2

Plant Area Assessment for Rainwater Harvesting

The topography, climate, soil, land use and hydrogeological conditions are important factors controlling the suitability of an area for ground water recharge. The prevalent topography, soil and land use conditions determine the extent of infiltration, whereas the hydrogeological conditions govern the occurrence of potential aquifer systems and their suitability for artificial recharge. The drainage system map of NSL Sugars Tungabhadra Unit is presented in Figure 13 below.

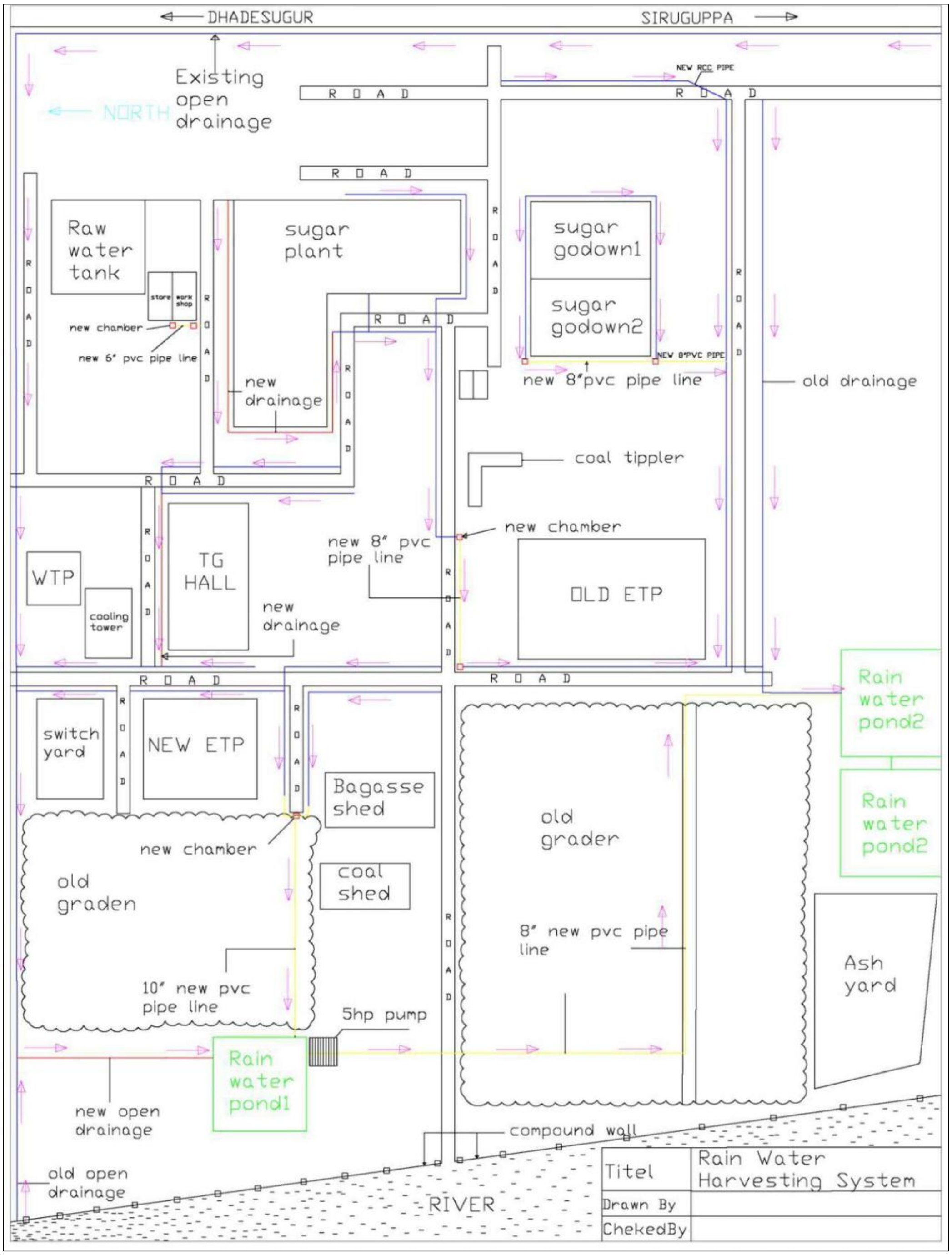


Figure 13: Drainage system map of NSL Sugars Tungabhadra Unit

The collection system includes the roof, gutters and downspouts of a building, and a network of pipes, drainages that collect the rainwater to a storage tank that is usually located to allow collection by gravity. The collection system also includes screens or traps to prevent debris from entering the storage tanks. A diversion device that prevents the first 0.5mm of rainwater in a rainfall event from entering the storage tanks i.e. first flush diverter, is also included. Total rooftop area of all the 8 major buildings considered is

17,490 m^2 as presented in Table 8 below and total surface area of catchment as per RWH report is **242,820 m^2**.

S. No.	Building Name	Rooftop Area in m^2
1	Sugar Plant	5232
2	TG Hall	1414
3	Bagasse Shed	1944
4	Coal Shed	1872
5	Store	1040
6	Workshop	572
7	WTP	416
8	Sugar Godowns	5000
	Total	**17,490**

Table 8: Total Rooftop Area of NSL Sugars Tungabhadra Unit

Different Catchments considered for rainwater harvesting is shown in Google Map as in Figure 14 below:

Figure 14: Different catchments identified in NSL Sugars Tungabhadra Unit

1. **Assessment of the Existing Strom water drains**

   Well-designed storm water systems are existing in the plant premises to make way for the runoff created due to rainfall, especially in Plant area. All the drains have been assessed based on the topography data provided by NSL Sugars and from the existing storm water drains, for which the carrying capacity of storm water generated in the plant premises is also analysed. In general, found that all the existing storm water drains have appropriate slopes for carrying runoff generated to the average rainfall of the area.

   The existing layout of the storm water drains in the premises is shown in Figure 13 with the direction of flow. It is also observed that some of the drains are being used for both the purpose of storm water and the processed waste water of various processes and which are not having enough storm water carrying capacity.

2. **Rain harvesting system design features outlined below are taken into account.**

   Rain fall data collection: Rain fall data collection on average rainfall over the last twelve years

   Determination of Catchment Area: The catchment area is the surface on which the rain water falls i.e. the collection of runoff water from land surfaces and factory building's roof tops which is **17,490 m^2** for building area and **242,820 m^2** for surface.

   First flushing device- Dispose-off the first spell of rain water so that it does not enter the system

   Filters - The filter is used to remove suspended pollutants from rainwater collected from rooftop water. The Various types of filters generally used for commercial purpose are Charcoal water filter, Sand filters, Horizontal roughing filter and slow sand filter.

   Storm water Conveyance system: Channels which surrounds edge of a sloping roof to collect and carryout rainwater to the storage tank. Gutters can be semi-circular or rectangular and mostly made locally from plain galvanized iron sheet. Conduits are pipelines or drains that carry rainwater from the catchment or rooftop area to the harvesting system. Commonly available conduits are made up of material like polyvinyl chloride (PVC) or galvanized iron (GI).

   Design of storm water Storage Pond - Depending on land space availability these tanks could be constructed above ground, partly underground or fully underground. Some maintenance measures like disinfection and cleaning are required to ensure the quality of water stored in the container.

   Recharge structures: Rainwater Harvested can also be used for charging the groundwater aquifers through suitable structures like dug wells, bore wells, recharge trenches and recharge pits.

   First flushing device: Roof washing or the collection and disposal of the first flush of water from a roof, is very important if the collected rain water is to be reused. All the debris, dirt and other contaminants especially bird dropping etc. accumulated on the roof during dry season are washed by the first rain and if this water will enter into storage tank it will contaminate the water. Therefore, to avoid this contamination a first flush system is incorporated in the roof top rain water harvesting system. The first flushing device, dispose-off the first spell of rain water so that it does not enter the system.

Sand filters: Rain water collected from roof top and ground surface, a filter unit is required to be installed in RWH system before storage tank. The filter is used to remove suspended pollutants from rain water collected over roof and ground surface. The filter unit is basically a chamber filled with filtering media such as fiber, coarse sand and gravel layers to remove debris and dirt from water before it enters the storage tank. The filter unit should be placed after first flush device but before storage tank

In sand filters, the primary filtering medium consists of commonly available sand, which is placed between two layers of gravel. The following figure 15 is the simple type of filter which is easy to construct and maintain. The sand fillers are very effective in removing turbidity and microorganism.

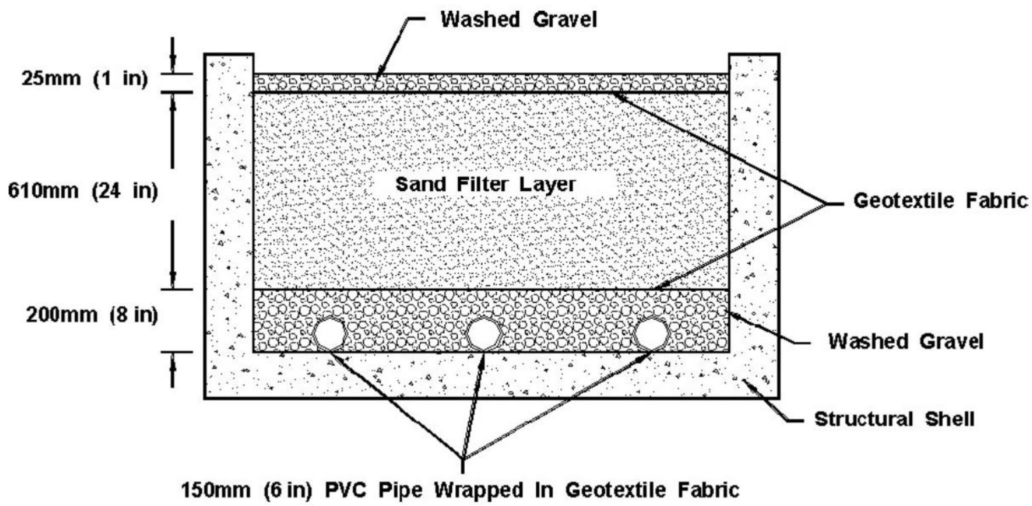


Figure 15: Sand Filter used in NSL Sugars Tungabhadra Unit

Storm water Conveyance system: Gutter is required to be used for collecting water from sloping roof and to divert it to downspout. The connection of gutters and down spouts should be done very carefully to avoid any leakage of water. The rain water collected on the roof top is transported down to storage facility through down spouts / conduits. Conduits can be of any material like PVC, GI or cast iron. The conduits should be free of lead and any other treatment which could contaminate the water. Coarse mesh / leaf screen to prevent the entry of leaves and other debris in the system, the coarse mesh should be provided at the mouth of inflow pipe. Storm water drains must be restored or rebuilt with an appropriate gradient to transport storm water flows from the catchment to the storage tanks as depicted in Figure 16 below.

Figure 16: Rainwater Storage Ponds in NSL Sugars Tungabhadra Unit

A.9. Implementation Benefits to Water Security

For Scope-5- Key Benefits:

a) Ensures compliance with environmental regulations by effectively reducing COD, BOD, and other pollutants in wastewater before discharge.
b) Protects local water bodies and groundwater resources from contamination, thereby preserving ecosystem health.
c) Promotes a circular economy through resource recovery in the form of treated water reuse, biogas generation, and bio-compost production.
d) Reduces freshwater consumption by reusing treated effluent and hot condensate water for process requirements.
e) Enhances water security by decreasing dependence on groundwater and external freshwater sources.
f) Improves resilience against water scarcity and ensures sustainable availability of water for industrial operations.
g) Minimizes wastewater generation and overall environmental footprint through efficient water management practices.

h) Improves energy efficiency by utilizing the thermal energy available in hot condensate, thereby reducing process heating requirements and operational costs.
i) Prevents seepage and contamination risks, thereby protecting soil quality and surrounding ecosystems.
j) Supports long-term sustainability, regulatory compliance, and climate-resilient industrial growth.

For Scope-2 Key Benefits:

a) Enables effective harvesting of rainwater from multiple hard and soft surface catchments across the plant area, maximizing utilization of rainfall.
b) Reduces surface runoff and prevents the loss of valuable rainwater resources.
c) Augments groundwater reserves through recharge, thereby enhancing long-term water security.
d) Reduces dependence on external water sources and minimizes stress on groundwater aquifers.
e) Improves resilience during lean periods and drought conditions by ensuring sustainable water availability.
f) Minimizes soil erosion, waterlogging, and localized flooding caused by excessive stormwater runoff.
g) Improves groundwater and nearby surface water quality through natural filtration and dilution processes.
h) Enhances soil moisture, supporting vegetation growth, greenbelt development, and ecological restoration.
i) Contributes to biodiversity conservation and strengthens overall ecological balance in the project area.
j) Improves catchment management and promotes sustainable utilization of water resources.
k) Strengthens climate resilience and supports sustainable water management practices for long-term environmental benefits.

A9.1 Objectives vs Outcomes

For Scope-5

Objective: The project aims to promote sustainable water resource management by treating and reusing wastewater generated from sugar and distillery operations, along with recovering and utilizing hot condensate water. The objective is to minimize freshwater withdrawal, reduce pollutant discharge, improve energy efficiency, enhance compliance with environmental regulations, and mitigate adverse impacts on surrounding ecosystems and communities through efficient water and waste management practices.

Outcome: The project has successfully established a circular water management approach wherein high-strength effluents are effectively treated and reused, resulting in substantial reductions in COD, BOD, and other contaminants before discharge. Reuse of treated wastewater and hot condensate has significantly reduced freshwater consumption and overall effluent generation while improving energy efficiency by utilizing the thermal energy available in condensate streams. The project has enhanced environmental performance, minimized odour and nuisance in the surrounding areas, reduced risks of soil and water contamination, and ensured compliance with applicable environmental standards. Collectively, these interventions have contributed to improved ecosystem health, sustainable water use, and a lower environmental footprint of the sugar and distillery operations.

For Scope-2

Objective: The project aims to conserve and sustainably utilize rainwater through the implementation of a scientifically designed rainwater harvesting system comprising multiple catchment areas. The objective is to augment groundwater resources, reduce stormwater runoff, improve water quality, strengthen water security, and enhance ecological balance by promoting efficient capture, storage, and recharge of rainwater.

Outcome: The project has effectively captured and utilized monsoon runoff, thereby enhancing groundwater recharge and improving the availability of water during lean periods. Reduced surface runoff has minimized soil erosion, waterlogging, and the risk of inundation in nearby areas, while also preventing pollutants from entering surrounding water bodies. The recharge process has contributed to improved groundwater quality through natural filtration and dilution mechanisms, supporting sustainable water availability and reducing dependence on external water sources. Increased soil moisture and improved catchment management have strengthened vegetation cover and ecological health, thereby supporting biodiversity and maintaining the overall environmental balance of the project area.

A9.2 Interventions by Project Owner / Proponent / Seller

For Scope-5

Interventions by Project Owner for ETP plant:

1. **Regular Monitoring and Maintenance:** Ensuring continuous monitoring of wastewater quality, system performance, and compliance with environmental standards through automated systems and regular inspections.
2. **Resource Recovery:** Implementing systems to recover biogas, treated water, or compost from the effluent treatment process, which adds value and reduces overall operational costs.
3. **Training and Skill Development**: Conducting regular training sessions for staff to ensure efficient operation and maintenance of the ETP, improving both performance and compliance with regulations.
4. **Compliance with Regulatory Standards**: Regularly updating the ETP to comply with evolving environmental regulations and local pollution control board requirements, ensuring sustainable operations.
5. **Public Awareness and Engagement:** Collaborating with local communities to raise awareness about the importance of ETP and its benefits for environmental protection and public health.
6. **Optimization of Chemical Usage:** Implementing strategies to optimize the use of chemicals in the treatment process, reducing operational costs and minimizing chemical waste.
7. **Zero Liquid Discharge (ZLD) Implementation:** Working towards a ZLD approach to ensure that no wastewater is discharged into the environment, promoting water reuse and reducing environmental impact.

Interventions by Project Owner for Use of Condensate water in plant:

1. **Infrastructure Upgrade:** Installed piping and storage systems to collect and transport hot condensate from process areas.
2. **Heat Recovery Systems:** Integrated heat exchangers to utilize thermal energy from hot condensate in preheating operations.

3. **Water Reuse Integration:** Modified existing processes to allow direct reuse of hot condensate as boiler feed water or for cleaning.
4. **Monitoring Systems:** Implemented flow meters and temperature sensors to monitor condensate recovery and reuse efficiency.
5. **Training & SOPs:** Conducted staff training and developed standard operating procedures for effective condensate management.
6. **Sustainability Alignment:** Ensured alignment of interventions with corporate water conservation and sustainability goals

For Scope-2

NSL Group identified huge potential and decided to take up the rainwater harvesting in project as discussed in above sections to conserve and store the rainwater and avoid the storm runoff.

1. **Installation of RWH Structures:** Project owners establish dedicated rainwater harvesting systems, such as recharge wells, ponds, and surface collection tanks, to capture and store rainwater during the monsoon season.
2. **Catchment Area Development:** They improve and maintain catchment areas, including roofs, paved surfaces, and surrounding land, to maximize rainwater collection potential.
3. **Filtration and Pre-treatment Systems:** Installing filtration units and first flush diverters to ensure clean and safe water storage, preventing contamination from debris or pollutants.
4. **Groundwater Recharge:** Implementing systems like recharge wells or pits to direct harvested rainwater into the ground, replenishing local aquifers and improving groundwater levels.
5. **Water Quality Monitoring:** Regular testing of stored rainwater to ensure its quality is suitable for reuse in plant operations or irrigation, meeting health and environmental standards.
6. **Reuse of Harvested Rainwater:** The harvested water is utilized for various plant needs, such as cooling systems, irrigation, cleaning, or non-potable uses, reducing dependence on external water sources.
7. **Monitoring and Optimization:** Using digital tools and sensors to monitor rainfall patterns, water levels, and system efficiency, enabling timely interventions and optimization of water collection.
8. **Sustainability Initiatives:** Aligning the rainwater harvesting strategy with broader sustainability goals, such as reducing the plant’s water footprint, conserving local water resources, and supporting the community’s water needs.

These interventions help to conserve water, reduce dependency on external sources, and promote sustainable water management within the sugar and distillery industry.

A.10. Feasibility Evaluation

For Scope-5

The installed ETP and condensate water recovery system by the project owner are robust and effectively handle variations in wastewater effluent. Prior to project implementation, a feasibility study was conducted in accordance with Karnataka State Pollution Control Board guidelines.

For Scope-2

The rainwater harvesting system installed by the project owner is robust and capable of managing heavy rainfall events. The catchment areas are well-graded to ensure efficient runoff collection, and the storage infrastructure is of high quality.

A.11. Ecological Aspects & Sustainable Development Goals (SDGs):

For Scope-5

This project demonstrates sustainable management and efficient utilization of water resources by treating and reusing ETP effluent and condensate water, thereby reducing dependence on groundwater extraction and conserving millions of liters of freshwater for distillery operations. The project also minimizes ecological impacts by preventing inundation of inhabited areas, reducing the risk of waterlogging and associated vector-borne diseases, and safeguarding groundwater quality through effective effluent treatment and the use of impervious infrastructure that prevents seepage and soil contamination. The initiative serves as a model for sustainable captive water management across water-intensive industries.

The ETP effectively treats the distillery effluent, and the use of impervious machinery within the ETP area further safeguards against potential leakage and contamination of surrounding soil.

For Scope-2

Rainwater harvesting is an environmentally sustainable practice that conserves water, enhances groundwater recharge, and supports ecosystem preservation. Its key ecological benefits include:

- Replenishes aquifers and reduces dependence on borewells and other groundwater sources, promoting sustainable water availability.
- Reduces surface runoff, thereby preventing soil erosion, inundation of inhabited areas, and waterlogging that can lead to vector-borne diseases.
- Improves groundwater and surface water quality by minimizing the entry of pollutants into water bodies.
- Supports aquatic ecosystems, biodiversity conservation, and the development of water-sensitive urban environments with minimal ecological disturbance.

SDGs Impacted- Linkage of the project activities with SDGs:

SDG 6 6 CLEAN WATER AND SANITATION	Clean Water and Sanitation – Enhances water availability, quality, and accessibility through harvesting, recharge, or conservation.
SDG 8 8 DECENT WORK AND ECONOMIC GROWTH	Decent Work and Economic Growth – Generates local employment during project implementation and maintenance.

SDG 12	Responsible Consumption and Production – Promotes sustainable use and management of natural resources.
SDG 13-	Climate Action – Contributes to climate resilience and adaptation by reducing water stress and improving local hydrological balance.
SDG 15	Life on Land – Supports ecosystem restoration, soil moisture retention, and biodiversity through improved water balance.

A.12. Recharge Aspects:

For Scope-5

Not Applicable, since this aspect does not lead to any recharging and storing facility for a long time for the water recycled.

For Scope-2

For the rainwater harvesting systems (RWH) indirectly contribute to recharging the bore wells within the project area and reduce the storm water flow in the area. First flushing device and sand filters and storm water conveyance system is implemented to prevent any contamination and maintain the quality of water before it enters the recharge areas. Consequently, the bore wells in the region are filled with water during the non-monsoon season, indicating successful groundwater recharge.

A.12.1 Solving for Recharge

For Scope-5

Water Budget Component	Typical Estimated Uncertainty (%)	Description
Surface Inflow	NA	NA
Precipitation	NA	NA
Surface Outflow	NA	NA
Evapotranspiration	NA	NA
Change in Storage	NA	NA
Deep Percolation	NA	NA

For Scope-2

Recharge = Rainfall + Surface Inflow – Evapotranspiration – Surface Outflow – Change in Storage

Evapotranspiration & Other Data: https://datameet-pune.github.io/open-water-data/docs/open-water-data-paper.pdf (or available under Documents Section- Water Data Guide)

Water Budget Component	Typical Estimated Uncertainty (%)	Description
Surface Inflow	1%	Typical range of accuracy from meters to minimum delivery accuracy requirements of delivery and diversion measurement devices.
Precipitation	2%	Typical range of accuracy from field-level rain gauges to extrapolation of local weather station data
Surface Outflow	1%	Typical range of accuracy from meters to estimated outflow relationships
Evapotranspiration	10%[4]	Clemmens and Burt, 1997; typical accuracy based on free water surface evaporation coefficient.
Change in Storage	5%	Estimated accuracy of change in storage calculation based on field scale water budget calibration to observed water levels.
Deep Percolation	5%	Typical range of calculated accuracy from field-scale water budget results (fields ranging from 56 to 125 acres)
Total	25%	

A.13. Quantification Tools

The following tools are recommended to be used to estimate the quantity of RoUs in the absence of tamperproof flow meters or systems that accurately quantify the volume in litres or m^3 of water being harvested or conserved by the project activity.

For Scope-5

The baseline scenario is the situation where, unutilized water is not conserved/treated/recycled and water is extracted from multiple bore wells within the project boundary which would have depleted the local groundwater resources (aquifers), in the absence of the project activity, the water for distillery process, gardening and other plant facilities usage would have been extracted from the ground water which have been avoided with the ETP implementation and reuse of condensate water installed within the project boundary.

[4] Since the harvested rainwater is utilized immediately and not stored for extended periods, and the flow distance within the project area is minimal, the potential for evaporation losses is significantly reduced. Therefore, a lower evaporation factor has been considered for the calculation than recommended.

Hence, the quantity of RoUs estimated as: *"the net quantity of treated wastewater that is gainfully used post treatment."*

Option -2 as per Rainwater Offset Standard version 8.1 is used and stated as:

Harvested/Recycled unutilized water or Volume of sea water utilized (m3) = Quantity of treated or recycled water in liters treated or harvested or volume of the artificial holding tank ($1m^3$ = 1000 liters).

The net quantity of treated water used is measured via flow meters installed at the site.

Further, as per *Rainwater Offset Standard version 8.1* a Conservative Approach: *The UWR RoU Verifier is recommended to apply a* ***10-50%*** *uncertainty factor related to degree of uncertainty to the final quantity of RoUs calculated for vintage years 2014-2021. However, a more conservative approach to uncertainty may be selected by the RoU Verifier as per its discretion.*

An uncertainty factor of 0.5% has been considered for the entire monitoring period from 01/01/2014 to 31/12/2025 for Scope 5 project. This factor reflects the high accuracy of the water flow measurements, which are obtained through calibrated flow meters with a standard accuracy of 0.5%. As these meters operate with minimal scope for manual intervention, the measurement process is free from human error and ensures consistent reliability throughout the monitoring period. Accordingly, the uncertainty factor has been derived based on this, representing a realistic and technically justified estimation.

The sugar ETP and CPU Unit data for the treated water and ROUs generated[5] is represented in the Table-9 below:

Year	Water treated in Distillery Plant (m3)	Water treated in Sugar plant (m3)
2014	42975	50434
2015	54288	34458
2016	41179	31466
2017	60957	47297
2018	16263	15690
2019	0	0
2020	0	0
2021	16797	15587
2022	14235	8338
2023	73796	42275

[5] For the calculations, please refer ER Sheet

2024	48122	32189
2025	29879	23275
ROUs	**368612**	**277734**
Total ROUs	**646,346**	
Final ROUs after uncertainty factor of 0.5%	**643,114**	

Table 9: Total RoUs for Scope-5 of NSL Sugars Tungabhadra Unit

For Scope-2

The baseline scenario is the situation where, unutilized water is not harvested into a well/aquifer/sub-surface structure/above-ground storage tank/collection chamber, in the absence of the project activity, the water for gardening and other plant facilities usage would have been extracted from the ground water which have been avoided with the rainwater harvesting method installed within the project boundary.

Hence, the quantity of RoUs is estimated as: *"the net quantity of rainwater harvested that is gainfully used in some other plant activity"*

Option -1 as per Rainwater Offset Standard is stated as:

Harvested/Recycled unutilized water or Volume of water utilized (m3) = Area of Catchment/Roof/Collection
Zone (m2) X Amount of rainfall (mm) X Runoff coefficient*Uncertainty Factor (1-0.25 = 0.75)

Further, as per *Rainwater Offset Standard version 8.1* a Conservative Approach: *The UWR RoU Verifier is recommended to apply a **10-50%** uncertainty factor related to degree of uncertainty to the final quantity of RoUs calculated for vintage years 2014-2021. However, a more conservative approach to uncertainty may be selected by the RoU Verifier as per its discretion.*

An uncertainty factor of 25% (0.75) has been considered for the entire monitoring period from 01/01/2014 to 31/12/2025 for Scope 2 project as per recharge aspect discussed in above section.

Water Harvesting Potential

Water harvesting potential of any catchment area is to be calculated under this methodology for each given year that the RoU is being claimed. The total amount of water that is received from rainfall over an area is called the rainwater legacy of that area. The amount that can be effectively harvested is called the water harvesting potential.

Assessment of Annual Rainwater Harvesting Potential (ARHP)

- Availability of source water is one of the basic prerequisites for taking up any rainwater harvesting and aquifer recharge scheme.
- The source of water available for aquifer recharge in the present case is only the rainfall-runoff which otherwise goes waste into local drains.

- A realistic assessment and quantification of the source water will help in designing the storage and recharge capacity of the RWH structures.
- The estimation of rainfall-runoff is made to quantify the water available for recharge, after all, losses.
- This estimated runoff will be used for the assessment of the size of the recharge structures.
- Though there are various methods to estimate the runoff, rational method is generally used as it is one of the most common and popularly used for small areas.
- The rational method is the sensible approach to determining the yield of a catchment by assuming the runoff coefficient.
- The **runoff coefficient** for various surfaces as per CGWA.
- Hence, based on each land use, its areas have been determined along with the corresponding runoff coefficient.
- The runoff coefficients established by CGWB[6], based on the field experiments are being used for the computation of rain water harvesting potential as listed in Table No. 10.
- Knowing the areas, runoff coefficient and average annual rainfall, the runoff from each land use has been determined. In this manner, all the land use, and open areas have been considered and cumulative runoff has been determined.
- The run-off coefficients for a different surface type are given in Table No. 10.

Different Surfaces	Runoff Coefficient (K)	Runoff Coefficient (K) used
Roof conventional (Flat)	0.7 to 0.8	0.7
Roof inclined (Sloping)	0.85 to 0.95	0.85
Concrete/Kota Paving	0.6 to 0.7	0.6
Gravel	0.5 to 0.7	0.5
Brick Paving	0.7	0.7

Table No. 10: Details of Runoff Coefficient (CGWA) and in line with the recommended Runoff Coefficient (K) by UWR ROU latest Standard

The following is the rational formula adopted to determine the yield of the catchment byconsidering the above runoff coefficient:

Annual rainwater harvesting potential is given by $V = K \times I \times A$

Where, V=Volume of water that can be harvested annually in litres.

K = Runoff coefficient

I = Annual rainfall in (mm)

[6] http://cgwa-noc.gov.in/LandingPage/index.htm

A = Catchment area in (m²

Based on the above method and using the land use pattern, estimated the Annual Rain Water Harvesting Potential (ARHP) for different areas. In this calculation of ARHP, the Average Annual Rainfall considered is **523.17 mm[7].** The table indicates that there is a tremendous scope to harvest the rain water to the tune of **82,626,327.5 l** within plant premises as indicated in Table 11.

Total Catchment	Type of Surface	Area Type	Gross Available Area in m2	Quantity of Runoff = V = K × I× A
	Landuse		**A**	**V (l)**
Rooftop Area	Roofs	Flat	17490	6405129.5
Surface	Surface	Concrete/Kota Paving	242820	76221198
	Total Area		**260310**	**82626327.5**

Table No. 11: Details of Annual Rain Water Harvesting Potential

Quantification

The formula for calculation for harvesting potential or volume of water received or runoff produced or harvesting capacity is given as:-

Harvesting potential or Volume of water utilized (m^3) = Area of Catchment/Roof/Collection Zone (m^2) X Amount of rainfall (m) X Runoff coefficient

Year	Total Runoff Generated		RoUs Generated	ROUs after applying Uncertainty factor of 0.75 (RoUs)
	Roofs	Surface		
2014	6881	81879	88759	66570
2015	7199	85667	92866	69649
2016	8986	106938	115924	86943
2017	6856	81588	88444	66333
2018	5485	65270	70755	53066
2019	2840	33801	36641	27481

[7] Please refer ER Sheet for calculation

2020	6170	73429	79599	59699
2021	5399	64250	69649	52237
2022	8693	103441	112134	84100
2023	4701	55946	60647	45485
2024	5118	60899	66017	49513
2025	8533	101547	110081	82561
Total	**76862**	**914654**	**991516**	**743637**

Table 12: Total RoUs for Scope-2 of NSL Sugars Tungabhadra Unit

Final ROUs from Both Scopes (Scope 2 & Scope 5) are as follows:

Final Vintage wise RoUs Calculation				
Year	**Water treated in Distillery (m3)**	**Water treated in Sugar plant (m3)**	**Total Runoff Generated (RoUs Generated)**	**Vintage wise ROU achieved**
2014	42760	50182	66570	159512
2015	54017	34286	69649	157952
2016	40973	31309	86943	159225
2017	60652	47061	66333	174045
2018	16182	15612	53066	84859
2019	0	0	27481	27481
2020	0	0	59699	59699
2021	16713	15509	52237	84459
2022	14164	8296	84100	106561
2023	73427	42064	45485	160976
2024	47881	32028	49513	129422
2025	29730	23159	82561	135449
Total	**366769**	**276345**	**661076**	**1304190**
Final ROU achieved in current period	**1304190**			

Table 13: Vintage wise Total RoUs from Scope 2 and Scope-5 of NSL Sugars Tungabhadra Unit

	Total RoUs generated with Uncertainty factor	Total RoUs generated without Uncertainty factor
Total RoUs Generated from TSL Unit	**1,304,190**	**1,527,781**
Annual RoUs Generated from TSL unit	**108,683**	**127,315**

A.14. UWR Rainwater Offset Do No Net Harm Principles

W.r.t Scope-5

The project activity accomplishes the following:

- Implementation of project activity prevents the release of wastewater directly into nearby river Tungabhadra which is the crucial step towards greener economy.
- The project activity does not harm any person or environment and it helps in improving the water quality of the area.
- The byproducts are also used as fertilizers by the farmers in nearby areas.
- This creates awareness among the employees and local people to practice such things on their level.

W.r.t Scope-2

The project activity accomplishes the following:

- Increases the sustainable water yield in areas where over development has depleted the aquifer and water level in surrounding areas.
- The catchments areas are built to collect the rainwater from going into storm drains or sewers and utilise in the facility for different purposes in future.
- This creates awareness among the employees and local people to practice such things on their level

A.15. Scaling Projects-Lessons Learned-Restarting Projects

W.r.t Scope-5

No Scope of extension observed as of now, by Project owner.

W.r.t Scope-2

No Scope of extension observed as of now, Project owner.